



# Characterizing the financial cycle: Evidence from a frequency domain analysis<sup>☆</sup>

Till Strohsal<sup>a,c,\*</sup>, Christian R. Proaño<sup>b</sup>, Jürgen Wolters<sup>a</sup>

<sup>a</sup> Freie Universität Berlin, Germany

<sup>b</sup> Otto-Friedrich-Universität Bamberg, Germany

<sup>c</sup> Federal Ministry for Economic Affairs and Energy

## ARTICLE INFO

### Article history:

Received 13 July 2016

Accepted 11 June 2019

Available online 14 June 2019

### JEL classification:

C22

E32

E44

### Keywords:

Financial cycle

Business cycle

Indirect spectrum estimation

Bootstrapping inference

## ABSTRACT

This paper introduces parametric spectrum estimation to the analysis of financial cycles. Our contribution is to formally test properties of financial cycles and to characterize their international interaction in the frequency domain. Existing work argues that the financial cycle is considerably longer in duration and larger in amplitude than the business cycle and that its distinguishing features became more pronounced over time. Also, a global cycle, being driven by US monetary policy, is said to be behind national financial cycles. We provide strong statistical evidence for the US and slightly weaker evidence for the UK validating the hypothesized features of the national financial cycle. In Germany, however, the financial cycle is much less visible. Similarly, a US-driven global financial cycle significantly affects national cycles in the UK but not in Germany.

© 2019 Elsevier B.V. All rights reserved.

## 1. Introduction

Fluctuations in financial markets play a key role in the macroeconomic dynamics of modern economies, often leading to either significant economic booms or severe economic crises, see e.g. Schularick and Taylor (2012) and Jordà et al. (2017). The last twenty years provide several examples, from the asset market crash in the early 1990s to the 2007/2008 global financial crisis which led the world economy to the brink of a new Great Depression. Against this background, a growing body of literature argues that the cyclical behavior of financial aggregates may be seen not only as a pure reflection of the real side of the economy, but also as the result of underlying changes in the general perception and attitudes towards financial and macroeconomic risk (see e.g. Borio, 2014 for a recent survey). For central banks and macroprudential regulation, the empirical and theoretical characterization of financial cycles has become one of the most central topics, see ECB (2014) and ESRB (2014).

The largest part of existing studies focuses on two aspects. The first aspect is the description of the empirical properties of finan-

cial cycles. According to Claessens et al. (2011), the most distinctive features of financial cycles seem to be that their duration is considerably longer and their amplitudes are larger than those of the classical business cycle. These results were obtained through a turning point analysis of financial aggregates using the dating algorithms of Bry and Boschan (1971) and Harding and Pagan (2002). Related studies apply frequency-based filters (Drehmann et al., 2012) and (Aikman et al., 2015), model-based filters (Koopman and Lucas, 2005), (Rünstler and Vlekke, 2018) and (Galati et al., 2016) or wavelet analysis (Verona, 2016) and (Ardila and Sornetto, 2016).<sup>1</sup> The extended length of the financial cycle is often considered to reflect the build-up of macro-financial instability which contributed most recently to the 2007/2008 financial crisis. Note, however, that the empirical properties of the financial cycle have not yet been formally tested.

The second aspect is the analysis of the synchronization of financial cycles across countries and the possible existence of a global financial cycle. Rey (2013) argues that the global financial cycle is highly correlated with the VIX and mainly driven by US monetary policy through the risk-taking channel. Using a large recursive VAR, she finds that the global financial cycle drives capital

<sup>☆</sup> The opinions in this paper are those of the authors and do not necessarily reflect the views of the German Federal Ministry for Economic Affairs and Energy.

\* Corresponding author.

E-mail address: [till.strohsal@fu-berlin.de](mailto:till.strohsal@fu-berlin.de) (T. Strohsal).

<sup>1</sup> For a Bayesian unobserved components approach see also Barbone Gonzalez et al. (2015).

flows, asset prices and credit growth around the world. The nexus of the financial cycle, monetary policy and the business cycle in the US is analyzed in Juselius et al. (2016). Passari and Rey (2015) document a transmission mechanism of US monetary policy shocks on the financial conditions in the United Kingdom. Further results at the global level from a Bayesian VAR are given by Miranda-Agrippino and Rey (2015). Empirical evidence for the risk-taking channel (Borio and Zhu, 2012) of US monetary policy at the domestic level is provided by Bekaert et al. (2013) and Bruno and Shin (2015). Common to most studies of the global financial cycle is that their results are based on a time domain analysis.

The contribution of this paper is twofold. First, as an alternative method to describe the properties of financial cycles in the frequency domain, we use the indirect spectrum estimation approach. A distinguishing feature of this method is its estimation efficiency. The estimated parameters and their standard errors allow for formal statistical testing of the cyclical properties of financial variables. We provide a thorough univariate frequency domain analysis of credit, the credit to GDP ratio, house prices and equity prices, all of which are considered key financial cycle indicators by existing work, see e.g. Claessens et al. (2011), Drehmann et al. (2012) and Borio (2014). Including data for the US, the UK and Germany, our study is structured around tests of the following hypotheses which we identify from the literature. a) Financial cycles are longer than business cycles. b) Financial cycles have a larger amplitude than business cycles. c) Financial cycles increased in length over time. d) Financial cycles are primarily a medium-term phenomenon. e) The properties of house price cycles and credit cycles are sufficiently similar to assume one financial cycle.

Second, we extend the literature by a multivariate frequency domain analysis of the interaction of financial cycles within countries and across countries. In contrast to an analysis in the time domain, our approach gives a more detailed picture and allows us not only to analyze whether there exists a relationship between financial variables at all, but also whether there is a relationship at specific frequencies. For instance, if there was a relationship only for a small range of frequencies it may not be discovered in the time domain. Similarly, if a significant relationship is found in the time domain, it remains unclear whether the interaction is taking place at high frequencies or - as one may suspect in an analysis of financial cycles - at low frequencies. Using a multivariate indirect spectrum estimation, we can formally test frequency-specific Granger-causality. Therefore, our multivariate frequency domain analysis complements and extends most recent time domain studies on the global financial cycle of, for instance, Rey (2013) and Miranda-Agrippino and Rey (2015). In particular, we investigate effects of US monetary policy through the risk-taking channel on financial and real activity in the UK and in Germany.

Our main findings are as follows. We find that the typical duration of the financial cycle, when measured by credit and house prices, has significantly increased in recent times in the United States and the United Kingdom, currently being about 15 years. Also, compared to the business cycle in these countries, we find strong statistical evidence in support of a longer duration and a larger amplitude of the financial cycle. In line with the results of Rünstler and Vlekke (2018), in the case of Germany, distinct characteristics of the financial cycle are, if at all, much less visible. A further clear result holds across all three countries: Equity has cyclical properties very similar to those of GDP and is hence mostly driven by cycles between 2 and 8 years. The multivariate analysis confirms the low frequency nature of credit and house prices. It also shows that these variables mainly interact at very low frequencies and that this interaction is statistically significant. As concerns the effect of US monetary policy through the risk-taking channel, we find significant Granger-causality at very low frequencies from the US VIX on credit and GDP in the UK.

For Germany, however, the evidence is again much weaker. The effect of the US VIX on Germany is not significant, but a tendency to a figure similar to that for the UK is visible. These findings add to our understanding of the cross-border impact of the risk-taking channel of US monetary policy. Since this channel operates mainly at medium-term frequencies, as our results indicate, and financial cycles have long-lasting effects, as argued also in Juselius et al. (2016), US monetary policy effects on the macroeconomic dynamics of other countries may last beyond the period of a business cycle.

The remainder of the paper is organized as follows. We discuss the methodology in Section 2. In Section 3, we present the data and the empirical results of our univariate analysis as well as the formal hypotheses tests. The multivariate analysis of the global financial cycle is shown in Section 4. Finally, we draw some conclusions from our study in Section 5.

## 2. Methodology

### 2.1. Univariate spectral analysis

The methodological starting point is the well-known fact that any covariance-stationary process has a time domain and a frequency domain representation which are fully equivalent, see Wiener (1930) and Khintchine (1934). The frequency domain representation of a time series is more suited to analyze its cyclical features. The spectrum of a process can, for instance, be used to derive the contribution of certain cycles to the overall variation of the process.<sup>2</sup> However, a (parsimonious) parametric time domain representation is better suited for estimation. Standard procedures can be used to consistently and efficiently estimate the parameters of a process. In order to combine the advantages of both representations, we apply the indirect spectrum estimation approach. In a nutshell, we do the estimation in the time domain and transform the estimated processes into the frequency domain to analyze their cyclical properties.

Our approach assumes that the data generating process (DGP) can be well approximated in the time domain by an autoregressive moving average (ARMA) model of the form

$$A(L)y_t = \delta + B(L)\varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma_\varepsilon^2), \quad (1)$$

where  $A(L)$  and  $B(L)$  denote polynomials in the lag operator  $L$  of order  $p$  and  $q$ , respectively. For stable processes, the  $\text{MA}(\infty)$  representation has the form

$$y_t - \mu = \frac{B(L)}{A(L)}\varepsilon_t, \quad \mu = \frac{\delta}{A(1)}. \quad (2)$$

Expanding the series  $B(L)/A(L)$  into a lag polynomial of order infinity shows that an ARMA representation can be interpreted as a filter of infinite length (which, however, depends only on a finite number of parameters). This specific filter transforms the white noise process  $\varepsilon_t$  into the observed time series  $y_t$ . As further detailed in B.1, the spectrum of  $y_t$  can be derived from the parameters of the process and is given by

$$f_y(\lambda) = \frac{|B(e^{-i\lambda})|^2}{|A(e^{-i\lambda})|^2} f_\varepsilon(\lambda), \quad (3)$$

where  $\lambda \in [-\pi, \pi]$  is the frequency,  $|B(e^{-i\lambda})|^2/|A(e^{-i\lambda})|^2$  represents the transfer function,  $f_\varepsilon(\lambda) = \frac{\sigma_\varepsilon^2}{2\pi}$  is the constant spectrum of the white noise process and  $i^2 = -1$ . Eq. (3) represents the main idea of the indirect spectrum estimation and allows us to derive

<sup>2</sup> This is because the spectrum represents an orthogonal decomposition of the variance.

the complete spectrum of  $y_t$  in a consistent and efficient way.<sup>3</sup> Normalizing the spectrum by the process variance yields the spectral density. The process variance is calculated by numerically integrating the spectrum. Our inference is based on bootstrapped standard errors; see B.4.

From  $f_y(\lambda)$  we can identify which frequency range explains most of the variance of  $y_t$ . The spectrum will exhibit a peak at a given frequency if the cyclical variation around that frequency is particularly important for the overall variation of the process. Also, if the spectral mass is more concentrated in a given range around the peak, the corresponding cycle will show a larger amplitude causing more regular swings in the time domain.

Note that a large part of the existing insights into the characteristics of the financial cycle is based either on the analysis of turning points (e.g. Claessens et al., 2011, 2012), frequency-based filter methods (e.g. Drehmann et al., 2012, Aikman et al., 2015) or on direct spectrum estimation (e.g. Schüler et al., 2015). Compared to our approach, the turning point approach requires a pre-specified rule which is applied to an observed time series in order to find local maxima and minima. Frequency-based filter methods require a pre-specified frequency range at which the financial cycle is assumed to operate. Directly estimating spectra requires very long time series to get precise estimates. For a more detailed comparison of our approach to filters and to non-parametric spectrum estimation, see Appendices B.1 and B.3.

## 2.2. Multivariate spectral analysis

In order to investigate the existence of a *common* financial cycle, we use the indirect spectrum estimation approach in a multivariate setup. This allows us to investigate at which frequencies a relationship between certain variables is particularly strong or weak. In order to keep the notation simple, consider the concrete case of a two-dimensional VAR model of order  $p$  with variable vector  $\mathbf{z}_t = (y_t, x_t)'$ ,

$$\mathbf{z}_t = A_1 \mathbf{z}_{t-1} + \dots + A_p \mathbf{z}_{t-p} + \mathbf{u}_t. \quad (4)$$

The  $(2 \times 2)$  coefficient matrices  $A_j$ ,  $j = 1, 2, \dots, p$  can be estimated consistently and efficiently by ordinary least squares. In a cointegrated VAR, ordinary least square still yields consistent estimates. The two-dimensional error vector  $\mathbf{u}_t = (u_y, u_x)'$  is assumed to be white noise with  $E(\mathbf{u}_t) = 0$  and a positive definite  $(2 \times 2)$  variance-covariance matrix  $E(\mathbf{u}_t \mathbf{u}_t') = \Sigma_u$ .

From the MA( $\infty$ ) representation  $\mathbf{z}_t = (I - A_1 L - \dots - A_p L^p)^{-1} \mathbf{u}_t$  we can derive the  $(2 \times 2)$  spectral matrix:

$$F_z(\lambda) = (I - A_1 e^{-i\lambda} - \dots - A_p e^{-ip\lambda})^{-1} \times \frac{\Sigma_u}{2\pi} (I - A_1 e^{-i\lambda} - \dots - A_p e^{-ip\lambda})^{-1'} \quad (5)$$

with  $0 \leq \lambda \leq \pi$  and  $(I - A_1 e^{-i\lambda} - \dots - A_p e^{-ip\lambda})^{-1'}$  denoting the complex conjugate.  $F_z(\lambda)$  includes the real valued spectra  $f_{xx}(\lambda)$  and  $f_{yy}(\lambda)$  and the complex valued cross-spectra  $f_{yx}(\lambda)$  and  $f_{xy}(\lambda) = f_{yx}(\lambda)$

$$F_z(\lambda) = \begin{pmatrix} f_{yy}(\lambda) & f_{yx}(\lambda) \\ f_{xy}(\lambda) & f_{xx}(\lambda) \end{pmatrix}. \quad (6)$$

The spectra  $f_{yy}(\lambda)$  and  $f_{xx}(\lambda)$  represent, just as in the univariate case, orthogonal decompositions of the variances of  $y$  and  $x$

in cyclical components. The spectral densities include information about the variance contributions of cycles at different frequencies.

The cross-spectra  $f_{yx}(\lambda)$  and  $f_{xy}(\lambda) = \overline{f_{yx}(\lambda)}$  can be used for measuring the strength of the relation between  $y$  and  $x$ . The squared coherency is given by

$$K_{yx}^2(\lambda) = \frac{|f_{yx}(\lambda)|^2}{f_{yy}(\lambda) f_{xx}(\lambda)} = K_{xy}^2(\lambda) \quad (7)$$

with  $0 \leq K_{yx}^2(\lambda) \leq 1$ . This measure is analogous to the coefficient of determination  $R^2$  for a linear relation between the cycles of  $y$  and  $x$  at frequency  $\lambda$ . Note that, in contrast to the  $R^2$ , the coherency is invariant to different linear transformations applied to  $y$  and  $x$ . In particular, this implies that it does not change when going from the relation in levels to the relation in first differences. Further, the coherency does not change when regressing  $y$  on  $x$  or  $x$  on  $y$ . Defining the  $(2 \times 2)$  matrix

$$B(L) = (I - A_1 L - \dots - A_p L^p)^{-1} P \quad (8)$$

with  $PP' = \Sigma_u$  and  $P$  as lower triangular regular matrix, the individual spectral densities of  $y$  and  $x$  can be written as

$$f_{yy}(\lambda) = \frac{1}{2\pi} (|B_{yy}(e^{-i\lambda})|^2 + |B_{yx}(e^{-i\lambda})|^2). \quad (9)$$

Based on this measure, Geweke (1982, 1984) and Breitung and Candelon (2006) proposed to test for Granger-causality at different frequencies using the statistic<sup>4</sup>

$$M_{x \rightarrow y}(\lambda) = \ln \left( 1 + \frac{|B_{yx}(e^{-i\lambda})|^2}{|B_{yy}(e^{-i\lambda})|^2} \right). \quad (10)$$

It holds that  $M_{x \rightarrow y}(\lambda) = 0$  if  $|B_{yx}(e^{-i\lambda})|^2 = 0$ , i.e., when  $x_t$  does not Granger-cause  $y_t$  at frequency  $\lambda$ . The higher the value of  $M_{x \rightarrow y}(\lambda)$  the stronger the impact of  $x$  on  $y$ .

## 2.3. Multivariate spectral analysis with cointegration

If the variables  $y_t$  and  $x_t$  in (4) are cointegrated, the system can be represented as a vector error correction model (VECM) where the relation between the first differences is separated from the long-run relation between the levels. However, cointegration also implies that the individual time series are non-stationary and thus their spectral densities have a pole at frequency  $\lambda = 0$ . Needless to say, the information that all variation of a trending variable is explained by a cycle of length infinity is only of limited use.

In contrast, the spectral densities of the first differences are more interesting. In practice, the spectral matrix of  $\Delta \mathbf{z}_t$  can be derived in two steps. The first step is to simply estimate the VAR in (4) under the parameter restriction that the variables are cointegrated

$$\mathbf{z}_t = A_1^r \mathbf{z}_{t-1} + \dots + A_p^r \mathbf{z}_{t-p} + \mathbf{u}_t. \quad (11)$$

Since a VECM requires to estimate less parameters than an unrestricted VAR, imposing cointegration on the VAR parameters in (4) increases estimation efficiency.

In the second step the multivariate first difference filter  $\Phi(e^{i\lambda})$  is applied to the spectral matrix of the levels. The frequency domain representation of the first difference vector time series is given by

$$F_{\Delta \mathbf{z}}(\lambda) = \Phi(e^{i\lambda}) F_z(\lambda) \Phi(e^{-i\lambda})'. \quad (12)$$

<sup>3</sup> As described in Hamilton (1994, p. 167), "If the model is correctly specified, the estimates [of the ARMA( $p, q$ ) model] will get closer and closer to the true values as the sample size grows; hence, the resulting estimate of the population spectrum should have this same property". The approach has also been applied by A'Hearn and Woitek (2001) in an analysis of the historical properties of international business cycles.

<sup>4</sup> For further details see e.g. Breitung and Schreiber (2017) where also a version of the test is discussed that allows for more detailed null hypotheses.

### 3. Frequency domain characterization of financial cycle variables

#### 3.1. Data description

According to Claessens et al. (2011), the core of financial intermediation should be well captured by the three distinct market segments of credit, housing and equity. Further financial variables considered in the literature are, for instance, the debt service ratio, see Juselius and Drehmann (2016). Drehmann et al. (2012) and Borio (2014) argue that the financial cycle can be most parsimoniously described in terms of credit and property prices, a result also found by Schuler et al. (2015) across European countries. Following these studies, in the univariate analysis of this section we use the most common proxy variables for the financial cycle, namely quarterly, seasonally adjusted aggregate data on credit volume, the credit to GDP ratio, house prices and equity prices. To study business cycles, we focus on real GDP. We analyze three industrialized countries: the United States (US), the United Kingdom (UK), and Germany. The US is not only the largest economy in the world, but it also has the worldwide most important financial sector. As representative countries for Europe, we consider the UK due to its leading financial sector and Germany as the largest EMU country.

We employ data transformations similar to those in Drehmann et al. (2012) to allow for a meaningful comparison to existing studies. All series are measured in logs, deflated with the consumer price index and normalized by their respective value in 1985Q1 to ensure comparability of units. Yearly growth rates are obtained by taking annual differences of each time series.<sup>5</sup> The only exception is the credit to GDP ratio which is expressed in percentage points and measured in deviations from a linear trend.<sup>6</sup> We use the longest possible sample for each individual time series which is mostly 1960Q1 until 2013Q4 for the US and the UK. Due to data availability the German time series start only in 1970Q1. The data are shown in Figs. 9–11 in Appendix A, where they are also described in more detail.

We split the data into two subsamples with break dates closely in line with the literature to analyze possible changes in the characteristics of the financial cycles over time. According to Claessens et al. (2011, 2012) and Drehmann et al. (2012), the break point is specified at 1985Q1 for the US and UK.<sup>7</sup> In the case of Germany, we choose the break point close to the German reunification, 1990Q2, because since then official data for the unified Germany are available. To statistically support the specified break dates which are adopted from the existing literature, we also estimate ARMA models for the full sample period and apply Chow break point tests as well impulse indicator saturation (IIS), an endogenous break point search; see Hendry (2011), Ericsson (2013) and the references therein. Chow tests provide statistical support for a break at 1985Q1 for the US and the UK. For Germany, the evidence for a break at 1990Q2 is weaker, but for GDP and credit we

clearly reject the null hypothesis of no break; see C.1. Nonetheless, the German reunification is a natural break date from an economic point of view. Also, the results from the IIS break point search indicate a break at mid to late 1980s for US and UK and around 1990 for Germany; see C.2. Finally, as it will be seen in the next subsection, the computed spectra are in most cases of substantially different shape across the subsamples. In Figs. 9–11 in Appendix A, the vertical gray line highlights the sample split. A first visual inspection of the data shows that the credit and house price growth rates exhibit more pronounced swings than GDP growth, particularly in the US and the UK. In Germany, this is only true for housing. In view of equity, the figures illustrate that these series not only feature a high volatility, but their dynamics seem to be very different from the other proxy variables for the financial cycle, credit and housing.

In the final part of the paper we use a number of additional variables, also sampled at the quarterly frequency. We use data on the VIX index, the Fed Funds rate as well as the \$/£ and \$/€ exchange rates.

#### 3.2. Univariate frequency domain characterization of financial cycle variables

We begin the discussion of our results by focusing on the spectral representation of the variables resulting from the estimated ARMA models. Further details on a concrete example of estimation and transformation are provided in B.5. As can be seen in C.1, all parameters in the final specifications are statistically significant at standard confidence levels and the estimated residuals are free from autocorrelation according to the Lagrange multiplier (LM) test. The specification of the ARMA models follows the principle of parsimony. Starting with the specification suggested by standard information criteria, we delete insignificant terms and include additional ones if necessary to guarantee the most important feature of a well-specified model: white noise residuals.

The estimated spectral densities are shown in Figs. 1–3 in the range  $[0, \pi/4]$ , i.e., for periods of  $\infty$  to 2 years.<sup>8</sup> We do not show frequencies in  $[\pi/4, \pi]$  since almost no spectral mass is located in this range. The traditional business cycle range of 8 to 2 years corresponds to the frequency interval  $[\pi/16, \pi/4]$ . The financial cycle range  $[\pi/64, \pi/16]$ , as proposed by Drehmann et al. (2012), includes cycles between 32 and 8 years and is highlighted by the gray area. An initial visual inspection of Figs. 1–3 delivers at least two noteworthy results.

First, especially for the US and the UK, the spectral densities of credit, credit to GDP and house prices are substantially shifted to the left in the later period compared to the first one. This indicates, at least superficially, that longer cycles could have become more present. Moreover, the peaks of the spectral densities of these variables are more pronounced, suggesting that longer cycles may have become more important for the variation of the process. For Germany, as illustrated in Fig. 3, this is only true for house prices. We obtain no clear results for German credit and credit to GDP. Therefore, it seems that German data do not provide much evidence in favor of the postulated financial cycle properties.

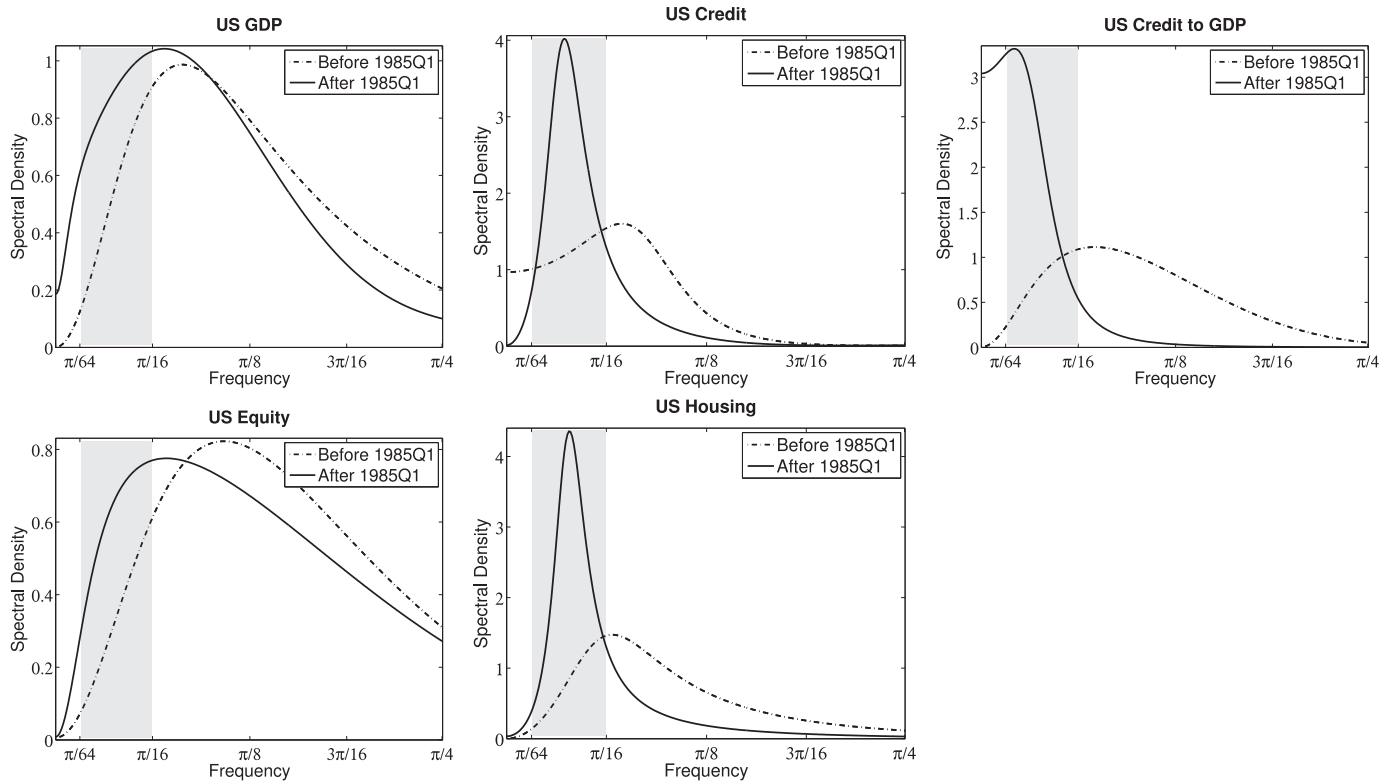
Second, Figs. 1–3 clearly show that in all three countries the spectral densities of GDP and equity changed very little from the period before the sample split to the one thereafter. The interesting exception is the spectrum of UK GDP growth which may indeed have experienced a change. The general impression, however, is that in almost all cases GDP and equity show very similar

<sup>5</sup> Our results from the univariate analysis therefore refer to cycles in yearly growth rates. The multivariate analysis in Section 3.3.5 shows, however, that the results are robust to using quarterly growth rates. B.2 provides a theoretical discussion of the frequency domain properties of the quarterly and yearly growth rate filters.

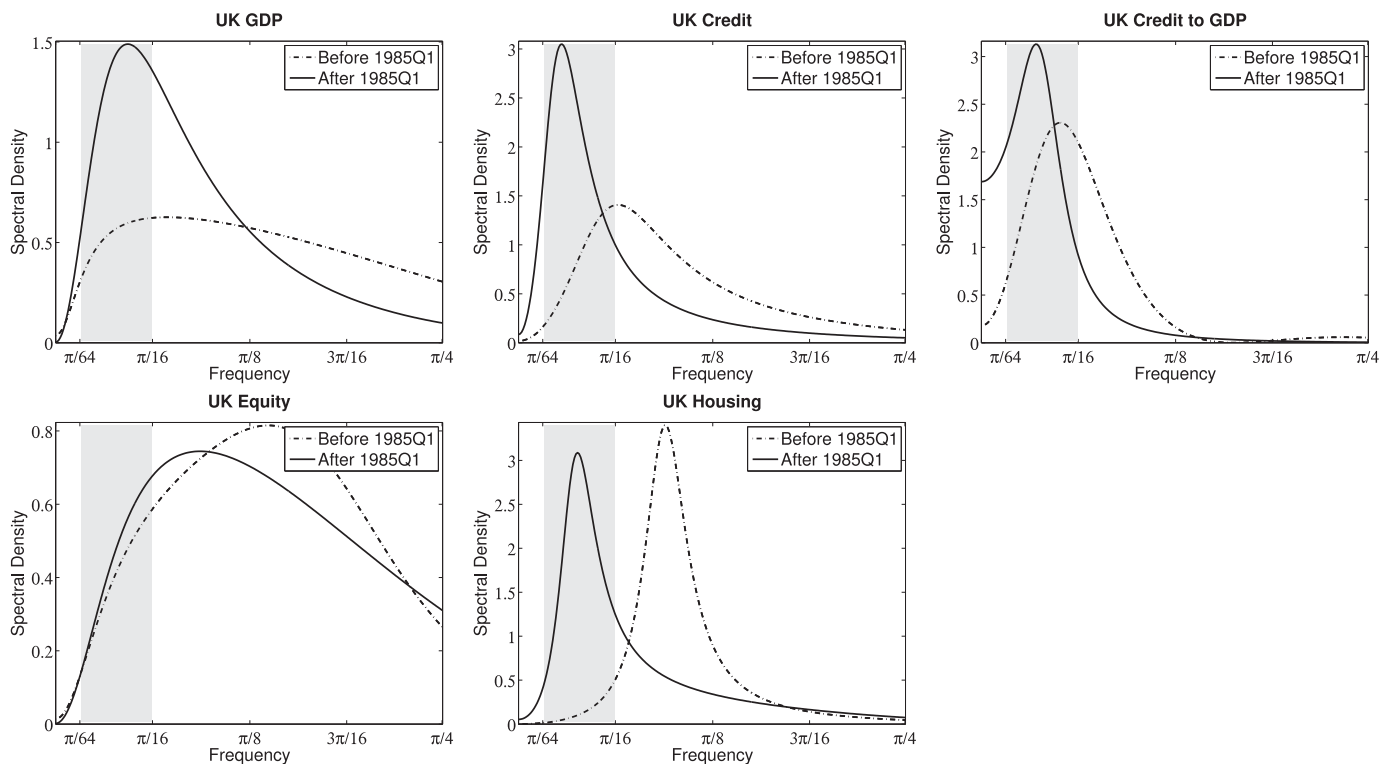
<sup>6</sup> Unit root tests of the level series indicate that, with the exception of the credit to GDP ratios which are found to be trend-stationary, all time series can be considered as integrated of order one. Therefore, working with annual growth rates for GDP, credit, housing and equity, i.e., annually differencing the data, is in line with the unit root test results. In the case of the credit to GDP ratio, we eliminated a deterministic linear trend. Results are available upon request.

<sup>7</sup> This is often seen as the starting point of the financial liberalization phase in mature economies. Moreover, during this period, monetary policy regimes being more successful in controlling inflation are established, see Borio (2014).

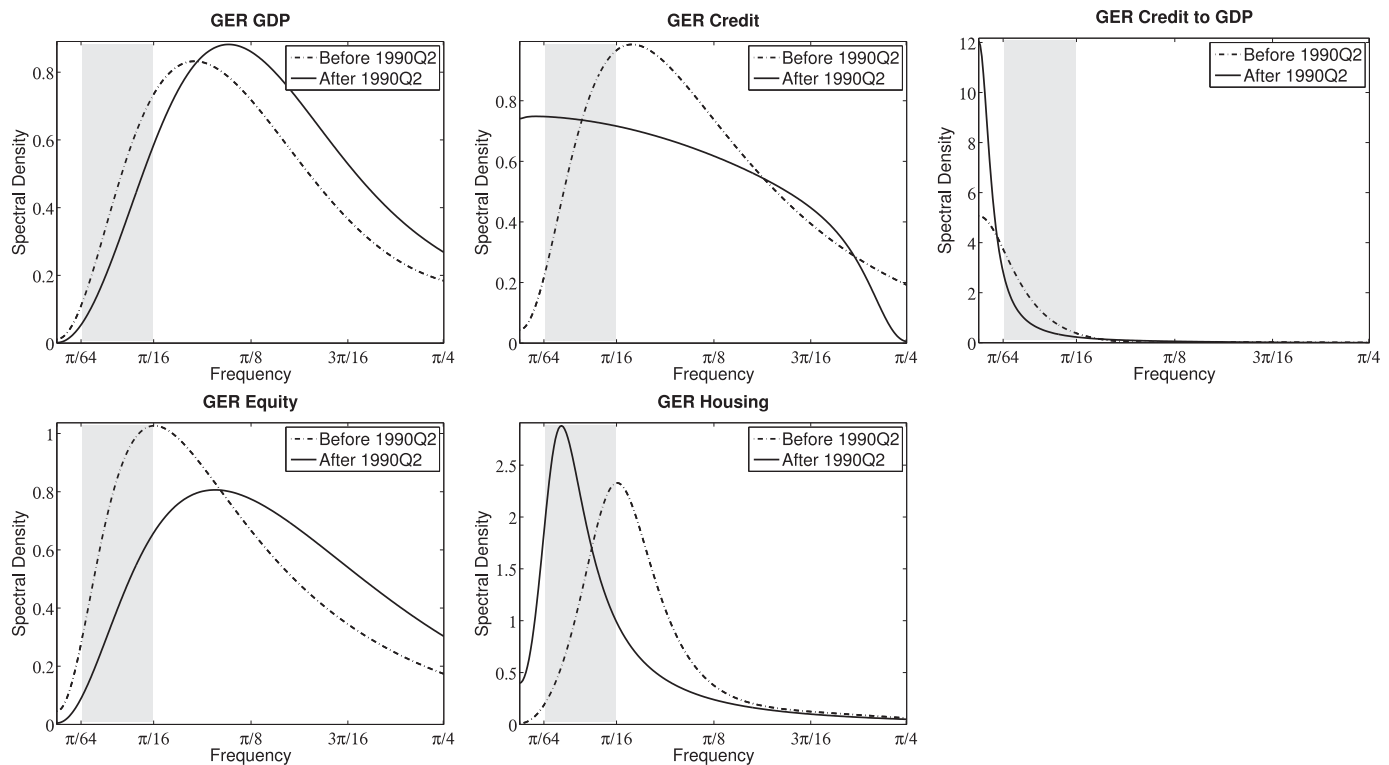
<sup>8</sup> When using quarterly data, all cycles of length infinity to half a year are described by the spectrum in the range from 0 to  $\pi$  because  $f_y(\lambda)$  is an even symmetric continuous function. We approximate the continuous spectrum by 1000 equally spaced frequency bands from 0 to  $\pi$ .



**Fig. 1.** Spectral Densities in the US. Note: Spectral densities in the gray area from frequencies  $\pi/64$  to  $\pi/16$  refer to the financial cycle range between 32 and 8 years, as proposed by Drehmann et al. (2012). Frequencies  $\pi/16$  to  $\pi/4$  correspond to cycles in the traditional business cycle range between 8 and 2 years.



**Fig. 2.** Spectral Densities in the UK. Note: Spectral densities in the gray area from frequencies  $\pi/64$  to  $\pi/16$  refer to the financial cycle range between 32 and 8 years, as proposed by Drehmann et al. (2012). Frequencies  $\pi/16$  to  $\pi/4$  correspond to cycles in the traditional business cycle range between 8 and 2 years.



**Fig. 3.** Spectral Densities in Germany. *Note:* Spectral densities in the gray area from frequencies  $\pi/64$  to  $\pi/16$  refer to the financial cycle range between 32 and 8 years, as proposed by [Drehmann et al. \(2012\)](#). Frequencies  $\pi/16$  to  $\pi/4$  correspond to cycles in the traditional business cycle range between 8 and 2 years.

spectral shapes with less pronounced peaks compared to the other variables.

It remains unclear from the graphical analysis, however, whether any of these properties changed significantly or whether certain variables have different properties. In preparation for the formal hypotheses testing in the next section, we propose four different statistics derived from the spectral densities: cycle length, amplitude, variance contribution of shorter cycles and variance contribution of longer cycles. We use these statistics to describe the main features of business and financial cycles.

[Table 1](#) summarizes the measures for all three countries under consideration and the two estimation subsamples. The first two columns include the *main cycle length* in years measured at the peak of the spectrum. It is given by  $\frac{2\pi}{\lambda_{\max}}$  with  $\lambda_{\max}$  as the frequency where the spectral density has its unique maximum. The remaining columns include the information about the *amplitude* and the variance contributions of certain frequency ranges. These statistics are related to the spectral mass, i.e. the area below the spectral density function, normalized to 1. For example, if the spectral mass between two frequencies, say  $[\lambda_{\text{low}}, \lambda_{\text{high}}]$ , would equal 0.5, we would conclude that 50% of the underlying time series' variance is explained by cycles in that frequency range. To approximate the variance contribution of the main cycle (the 'amplitude'), we report the spectral mass, measured in percentage points, which is located around  $\lambda_{\max}$ . We choose a symmetric frequency band with length of about  $\frac{\pi}{20}$ .<sup>9</sup> We also calculate the spectral mass for pre-defined ranges which are often used in the literature, i.e., 8 to 32 years for the financial cycle and 2 to 8 years for the business

cycle. The values in brackets below the point estimates are the 95% bootstrap confidence intervals.

According to the point estimates in [Table 1](#), the duration of the financial cycle may have increased from the first to the second subsample. Averaging over the four financial variables, we find cycles of 6.5, 6.6 and 7.6 years during the first period, and 14.4, 12.7 and 11.7 years during the second period in the US, UK, and Germany, respectively.<sup>10</sup> However, our estimates do not show much of a change in the length of the business cycle. This would be in line with the widespread notion that business cycles have a duration between 2 and 8 years, see [Hodrick and Prescott \(1997\)](#).

[Rünstler and Vlekke \(2018\)](#) find business cycles to be even longer than 8 years. Applying a multivariate unobserved components model to *level* data, they estimate a GDP cycle with a duration of 8.7 years for the US and 13.4 years for the UK.<sup>11</sup> The business cycles shown in [Table 1](#) refer to yearly growth rates and have a shorter duration. One interpretation of the different results is that medium-term cycles, if present in the *level* of GDP, get attenuated in our analysis by calculating the percentage differences to last year's quarter, because this is equivalent to applying a specific difference filter, see [B.2](#). Therefore, we emphasize that the present paper focuses on the cyclical properties of yearly *growth rates*.

<sup>9</sup> Using a band is necessary because the variance contribution of single frequency is zero by definition. The band is defined as  $\lambda_{\max} \pm \frac{\pi}{1000} \cdot 25$  and its size therefore equals 5% of all frequencies. In that sense it is referred to as "small". During computation we discretize the frequencies into 1000 steps.

<sup>10</sup> For Germany, cases where we do not obtain finite standard errors are not taken into account. For instance, we find a main cycle estimate of 45.4 years with unbounded standard errors for German credit in the second subsample. In order to study whether this actually reflects a unit root or only a break in the deterministic part we conducted unit root tests with endogenous break point search. However, the test results were inconclusive. Since the interpretation of the main cycle estimate for German credit is difficult and the estimation uncertainty is large, we excluded it from subsequent hypotheses testing.

<sup>11</sup> [Beaudry et al. \(2016\)](#) find the main cycle in the levels of many important US macroeconomic variables such as hours worked, unemployment and capacity utilization to be around 10 years; see also [de Winter et al. \(2017\)](#).

**Table 1**  
Frequency domain: Length and variance contribution of cycles.

|     |               | Spectral mass in percent located... |                      |                                          |                      |                                             |                      |                                             |                      |
|-----|---------------|-------------------------------------|----------------------|------------------------------------------|----------------------|---------------------------------------------|----------------------|---------------------------------------------|----------------------|
|     |               | Length of main cycle<br>in years    |                      | ...at the amplitude<br>of the main cycle |                      | ...at longer-term<br>cycles (8 to 32 years) |                      | ...at shorter-term<br>cycles (2 to 8 years) |                      |
|     |               | Pre                                 | Post                 | Pre                                      | Post                 | Pre                                         | Post                 | Pre                                         | Post                 |
| US  | GDP           | 6.2<br>[4.6, 7.2]                   | 7.1<br>[4.9, 17.2]   | 29.9<br>[22.3, 35.8]                     | 31.9<br>[27.2, 42.3] | 17.2<br>[9.9, 23.1]                         | 26.8<br>[10.9, 39.4] | 71.4<br>[66.6, 74.9]                        | 63.3<br>[45.1, 77.7] |
|     | Credit        | 6.9<br>[5.7, 12.2]                  | 13.9<br>[12.8, 14.7] | 46.1<br>[39.1, 59.7]                     | 75.4<br>[71.9, 78.6] | 37.8<br>[16.9, 48.6]                        | 76.2<br>[71.2, 79.2] | 51.4<br>[34.9, 80.6]                        | 19.3<br>[15.4, 23.3] |
|     | Credit to GDP | 6.8<br>[5.1, 9.4]                   | 23.8<br>[14.7, 45.4] | 33.7<br>[27.8, 36.7]                     | 86.4<br>[75.2, 91.2] | 23.8<br>[13.6, 31.2]                        | 61.8<br>[49.0, 71.6] | 66.0<br>[54.7, 71.0]                        | 6.8<br>[4.1, 12.8]   |
|     | Housing       | 7.6<br>[4.0, 9.6]                   | 12.8<br>[10.2, 14.7] | 41.4<br>[20.6, 72.3]                     | 73.0<br>[61.4, 81.0] | 26.0<br>[3.3, 37.7]                         | 72.6<br>[55.7, 80.6] | 66.2<br>[53.5, 89.9]                        | 24.9<br>[17.5, 41.3] |
|     | Equity        | 4.6<br>[3.3, 5.6]                   | 7.0<br>[5.4, 8.5]    | 25.3<br>[21.7, 30.9]                     | 23.9<br>[21.0, 27.1] | 10.6<br>[3.9, 15.0]                         | 19.0<br>[14.1, 22.3] | 74.2<br>[67.5, 82.0]                        | 65.4<br>[61.6, 69.3] |
|     |               |                                     |                      |                                          |                      |                                             |                      |                                             |                      |
| UK  | GDP           | 6.9<br>[6.2, 8.1]                   | 10.9<br>[8.2, 12.5]  | 19.4<br>[19.3, 19.5]                     | 41.8<br>[32.0, 47.0] | 16.3<br>[14.9, 17.4]                        | 38.3<br>[25.8, 45.5] | 58.6<br>[57.3, 59.6]                        | 55.8<br>[48.3, 65.2] |
|     | Credit        | 7.8<br>[4.6, 9.4]                   | 17.8<br>[15.1, 20.0] | 39.5<br>[21.4, 54.3]                     | 60.6<br>[53.7, 64.9] | 26.2<br>[8.1, 38.5]                         | 62.4<br>[55.1, 65.5] | 63.5<br>[53.5, 77.3]                        | 27.6<br>[23.0, 35.3] |
|     | Credit to GDP | 9.8<br>[5.7, 13.9]                  | 14.3<br>[8.8, 31.2]  | 62.1<br>[46.5, 79.4]                     | 73.2<br>[52.0, 87.1] | 54.5<br>[18.2, 63.2]                        | 68.3<br>[42.6, 83.4] | 39.7<br>[29.5, 76.6]                        | 13.1<br>[6.3, 41.3]  |
|     | Housing       | 5.3<br>[3.7, 6.4]                   | 13.1<br>[11.1, 14.3] | 69.2<br>[35.0, 96.8]                     | 58.1<br>[33.5, 72.2] | 5.1<br>[0.3, 16.6]                          | 58.2<br>[30.0, 72.7] | 93.0<br>[76.8, 99.4]                        | 37.8<br>[24.9, 63.0] |
|     | Equity        | 3.6<br>[2.4, 6.4]                   | 5.4<br>[3.3, 6.8]    | 25.2<br>[20.7, 47.9]                     | 23.0<br>[20.6, 27.1] | 12.1<br>[2.2, 19.5]                         | 14.0<br>[4.3, 18.1]  | 75.3<br>[61.5, 89.5]                        | 68.4<br>[62.8, 77.2] |
|     |               |                                     |                      |                                          |                      |                                             |                      |                                             |                      |
| GER | GDP           | 5.6<br>[5.0, 7.8]                   | 4.5<br>[2.9, 5.7]    | 25.4<br>[20.7, 32.1]                     | 27.0<br>[21.7, 33.1] | 13.8<br>[11.2, 23.2]                        | 9.4<br>[3.0, 15.4]   | 62.2<br>[58.0, 62.8]                        | 74.2<br>[66.0, 81.6] |
|     | Credit        | 6.9<br>[6.1, 8.5]                   | 45.4<br>[–, –]       | 29.8<br>[25.2, 36.9]                     | 23.3<br>[–, –]       | 20.7<br>[16.5, 28.6]                        | 22.2<br>[–, –]       | 67.9<br>[63.6, 69.3]                        | 56.6<br>[–, –]       |
|     | Credit to GDP | ∞<br>[–, –]                         | ∞<br>[–, –]          | 86.9<br>[69.7, 94.5]                     | 91.4<br>[45.2, 96.5] | 45.6<br>[27.7, 51.8]                        | 23.5<br>[10.1, 54.4] | 4.1<br>[1.7, 10.4]                          | 5.7<br>[2.2, 41.2]   |
|     | Housing       | 7.9<br>[4.9, 12.8]                  | 18.5<br>[11.6, 20.8] | 59.8<br>[33.2, 72.8]                     | 60.6<br>[35.8, 70.3] | 39.3<br>[11.2, 55.8]                        | 60.4<br>[33.2, 68.7] | 59.4<br>[38.7, 84.4]                        | 27.7<br>[19.6, 55.0] |
|     | Equity        | 7.9<br>[4.8, 10.2]                  | 4.9<br>[3.0, 6.4]    | 30.7<br>[18.6, 40.7]                     | 24.9<br>[21.5, 31.4] | 23.78<br>[10.4, 32.9]                       | 12.0<br>[2.9, 17.4]  | 63.0<br>[56.0, 71.1]                        | 72.5<br>[65.3, 82.3] |
|     |               |                                     |                      |                                          |                      |                                             |                      |                                             |                      |

Notes: The terms *pre* and *post* refer to the sample periods 1960Q1 until 1984Q4 and 1985Q1 until 2013Q4, respectively, for the US and the UK. In the case of Germany the samples are 1970Q1 until 1990Q1 and 1990Q2 until 2013Q4. The length of the main cycle is defined as  $\frac{2\pi}{\lambda_{\max}}$ , where  $\lambda_{\max}$  is the frequency where the spectral density has its unique maximum. The amplitude of the main cycle is defined as the spectral mass in a “small” frequency band symmetric around  $\lambda_{\max}$  with a length of about  $\frac{\pi}{20}$ . 95% bootstrap confidence intervals are given in brackets. By “–” it is indicated that there is no distinct solution (maximum and percentiles) in the frequency range  $0 < \lambda \leq \pi$ . Accordingly, a main cycle length of “∞” is sometimes obtained because all spectral mass is located at frequency zero, pointing to a unit root.

### 3.3. Hypotheses tests

Let us now see which of the above impressions survive statistical testing. We investigate five distinct hypotheses. a) The financial cycle is longer than the business cycle. b) The financial cycle increased in length over time. c) The financial cycle has a larger amplitude than the business cycle. d) The importance of the financial cycle increase over time. e) The properties of house price cycles and credit cycles are sufficiently similar to assume one financial cycle.

#### 3.3.1. Hypothesis a: The financial cycle is longer than the business cycle

As a starting point, we test whether the financial cycle tends to be a medium-term phenomenon with a significantly longer cycle length than that of the business cycle as suggested by Claessens et al. (2011), Drehmann et al. (2012) and Borio (2014), among others. Table 2 reports the test results of the null hypothesis that the financial cycle and the business cycle are of equal length against the alternative hypothesis that the financial cycle is longer than the business cycle. According to the one-sided two-sample *t*-tests, the null hypothesis cannot be rejected in the first subsample for any of the countries; at least at the 5%-level. This holds also for the second subsample for the credit to GDP ratio

and equity. The cycle lengths of credit and housing, however, are significantly longer than the one of GDP. With respect to the information given in Table 1 this is not surprising. In the second subsample the average financial cycle is twice as long as the average business cycle.

#### 3.3.2. Hypothesis b: The financial cycle has a larger amplitude than the business cycle

Next, we test whether the financial cycle and the business cycle feature the same amplitude in the two subperiods. Table 3 shows a clear result for equity, where the amplitude is never found to be different from the one of GDP. For the other variables we get somewhat mixed evidence in the first sample period, as the null is rejected only in a few cases. In the second period, however, the results are again very clear. Apart from equity, we strongly reject for all variables in all countries. This implies that, particularly in recent times and when taking credit, the credit to GDP ratio and housing as indicators, the financial cycle is characterized by a significantly larger amplitude than the business cycle.

#### 3.3.3. Hypothesis c: The financial cycle increased in length over time

We now analyze whether the medium-term nature of the financial cycle is a recent phenomenon, i.e., whether its length increased over time. Statistical evidence is shown in Table 4.

**Table 2**

Is the financial cycle longer than the business cycle?

$H_0$ : The financial cycle and the business cycle are of equal length.  
 $H_1$ : The financial cycle is longer than the business cycle.

|     |               | Pre break       |         | Post break      |         |
|-----|---------------|-----------------|---------|-----------------|---------|
|     |               | $\hat{t}$ -stat | p-value | $\hat{t}$ -stat | p-value |
| US  | Credit        | 0.26            | 0.399   | 2.45            | 0.007   |
|     | Credit to GDP | 0.47            | 0.319   | 0.64            | 0.260   |
|     | Housing       | 0.76            | 0.224   | 1.95            | 0.025   |
|     | Equity        | −1.57           | 0.942   | −0.04           | 0.514   |
|     |               |                 |         |                 |         |
| UK  | Credit        | 0.64            | 0.259   | 4.28            | 0.000   |
|     | Credit to GDP | 1.19            | 0.118   | 0.26            | 0.399   |
|     | Housing       | −1.86           | 0.968   | 1.72            | 0.042   |
|     | Equity        | −1.82           | 0.966   | −3.63           | 0.999   |
|     |               |                 |         |                 |         |
| GER | Credit        | 1.37            | 0.085   | –               | –       |
|     | Credit to GDP | –               | –       | –               | –       |
|     | Housing       | 0.90            | 0.185   | 5.39            | 0.000   |
|     | Equity        | 1.40            | 0.080   | 0.34            | 0.366   |
|     |               |                 |         |                 |         |

Notes:  $\hat{t}$ -stat represents the estimated value of the  $t$ -statistic of a one-sided two-sample  $t$ -test. By “–” it is indicated that we could not obtain finite bootstrap standard deviations which implies that it is not possible to conduct a  $t$ -test.

**Table 3**

Does the financial cycle have a larger amplitude than the business cycle?

$H_0$ : The financial cycle and the business cycle have the same amplitude.  
 $H_1$ : The financial cycle has a larger amplitude than the business cycle.

|     |               | Pre break       |         | Post break      |         |
|-----|---------------|-----------------|---------|-----------------|---------|
|     |               | $\hat{t}$ -stat | p-value | $\hat{t}$ -stat | p-value |
| US  | Credit        | 2.64            | 0.004   | 6.48            | 0.000   |
|     | Credit to GDP | 0.92            | 0.178   | 7.13            | 0.000   |
|     | Housing       | 0.82            | 0.207   | 5.05            | 0.000   |
|     | Equity        | −1.09           | 0.861   | −1.19           | 0.884   |
|     |               |                 |         |                 |         |
| UK  | Credit        | 2.27            | 0.012   | 3.97            | 0.000   |
|     | Credit to GDP | 5.01            | 0.000   | 3.19            | 0.000   |
|     | Housing       | 3.01            | 0.001   | 1.66            | 0.049   |
|     | Equity        | 0.77            | 0.219   | −4.36           | 0.999   |
|     |               |                 |         |                 |         |
| GER | Credit        | 0.96            | 0.169   | –               | –       |
|     | Credit to GDP | 8.29            | 0.000   | 4.80            | 0.000   |
|     | Housing       | 3.17            | 0.001   | 3.54            | 0.000   |
|     | Equity        | 0.81            | 0.209   | −0.52           | 0.698   |
|     |               |                 |         |                 |         |

Notes: The approximate amplitude is defined as the spectral mass in the symmetric frequency band with a length of about  $\frac{\pi}{20}$  around  $\lambda_{\max}$ , where  $\lambda_{\max}$  is the frequency where the spectral density has its unique maximum.  $\hat{t}$ -stat represents the estimated value of the  $t$ -statistic of a one-sided two-sample  $t$ -test. By “–” it is indicated that we could not obtain finite bootstrap standard deviations which implies that it is not possible to conduct a  $t$ -test.

The results clearly support the hypothesis for credit and housing, both having longer cycles during the second sample period. These results thus provide statistical support for the findings of Claessens et al. (2011), Drehmann et al. (2012) and Borio (2014). However, the tests also show that it should be looked closely at each variable. For the credit to GDP ratio and equity we find almost no evidence for an increase in the cycle length.

### 3.3.4. Hypothesis d: The importance of the financial cycle increased over time

Next, we investigate whether the financial cycle has become more important over time. We address this question by testing whether the main cycles amplitude has increased, i.e. whether a larger part of the overall process variance is explained by the financial cycle. A higher amplitude would be reflected by a higher peak of the main cycle, implying an increase in its variance contribution. As can be seen from Table 5, the  $t$ -tests strongly support the alternative hypothesis of a significant increase for US credit, credit to GDP and housing. In the UK, the  $t$ -test results deliver a

**Table 4**

Did the financial cycle increase in length over time?

$H_0$ : The length of the financial cycle has not changed over time.  
 $H_1$ : The length of the financial cycle has increased over time.

|     |               | $\hat{t}$ -stat | p-value |
|-----|---------------|-----------------|---------|
| US  | Credit        | 2.32            | 0.010   |
|     | Credit to GDP | 0.66            | 0.256   |
|     | Housing       | 2.59            | 0.005   |
|     | Equity        | 2.35            | 0.009   |
|     |               |                 |         |
| UK  | Credit        | 5.69            | 0.000   |
|     | Credit to GDP | 0.33            | 0.370   |
|     | Housing       | 7.26            | 0.000   |
|     | Equity        | 0.84            | 0.200   |
|     |               |                 |         |
| GER | Credit        | –               | –       |
|     | Credit to GDP | –               | –       |
|     | Housing       | 3.01            | 0.001   |
|     | Equity        | −1.74           | 0.959   |
|     |               |                 |         |

Notes:  $\hat{t}$ -stat represents the estimated value of the  $t$ -statistic of a one-sided two-sample  $t$ -test that compares the cycle lengths in the pre and post sample period. By “–” it is indicated that we could not obtain finite bootstrap standard deviations which implies that it is not possible to conduct a  $t$ -test.

**Table 5**

Did the importance of the financial cycle increase over time?

$H_0$ : The main cycle's amplitude has not changed over time.  
 $H_1$ : The main cycle's amplitude has increased over time.

|     |               | $\hat{t}$ -stat | p-value |
|-----|---------------|-----------------|---------|
| US  | Credit        | 5.38            | 0.000   |
|     | Credit to GDP | 11.06           | 0.000   |
|     | Housing       | 2.18            | 0.015   |
|     | Equity        | −0.49           | 0.688   |
|     |               |                 |         |
| UK  | Credit        | 2.28            | 0.011   |
|     | Credit to GDP | 0.89            | 0.186   |
|     | Housing       | −0.59           | 0.722   |
|     | Equity        | −0.28           | 0.611   |
|     |               |                 |         |
| GER | Credit        | –               | –       |
|     | Credit to GDP | 0.31            | 0.379   |
|     | Housing       | 0.06            | 0.478   |
|     | Equity        | −0.93           | 0.825   |
|     |               |                 |         |

Notes: The approximate amplitude is defined as the spectral mass in the symmetric frequency band with a length of about  $\frac{\pi}{20}$  around  $\lambda_{\max}$ , where  $\lambda_{\max}$  is the frequency where the spectral density has its unique maximum.  $\hat{t}$ -stat represents the estimated value of the  $t$ -statistic of a one-sided two-sample  $t$ -test. By “–” it is indicated that we could not obtain finite bootstrap standard deviations which implies that it is not possible to conduct a  $t$ -test.

significant result only for the credit series. Germany seems again to be characterized by different dynamics, as the main cycles' amplitudes of the credit to GDP ratio as well as the growth rates of house prices and equity do not appear to have changed between the two time periods.

One explanation of the different findings for Germany is given by Rünstler and Vlekke (2018), who point out that the private home ownership rate is considerably lower in Germany than in the US and the UK. This feature may have shielded the German housing market from the effects resulting from the financial globalization of the last three decades and the related higher availability of financing capital. Additionally, while we find long cycles in German house prices, they have not become more important over time. Also, the cycles in credit and credit to GDP are not as pronounced as in the US and the UK.

Altogether, the results show a statistically significant change of the overall shape of the spectral density over time, particularly for credit and housing in the US and the UK. Further, the largest share of spectral mass of the credit and housing series in

the US and UK is clearly located at cycles longer than the business cycle. Therefore, our results also support the conjecture of Drehmann et al. (2012) and Borio (2014) that particularly credit and house prices capture the most important features of financial cycles. Credit represents a direct financing constraint to the economy and real estate is one of the assets that is mainly financed by credit. Besides, house prices can be related to the perception of value and risk in the economy. For a detailed discussion of the role of subjective and possibly biased perceptions of value and risk in the housing market, see Geanakoplos et al. (2012).

Our results also highlight that credit and house prices should be at the center of interest when monitoring the financial cycle. To some extent, this is already the case. With the intention of curbing the financial cycle, the European Systemic Risk Board recommends the credit to GDP ratio as financial cycle indicator to guide the setting of counter-cyclical buffer rates, see ESRB (2014). However, our results indicate that the credit to GDP ratio shows only a few of the hypothesized financial cycle properties. This implies that there is considerable statistical uncertainty associated with the estimation of the frequency domain representation of that time series. It seems that when credit and GDP are considered individually, they are quite informative, but when forming the ratio, the combined measure provides only a mixed picture.

A formal interpretation of that finding is suggested by the representation of credit and GDP in terms of their  $p$  reciprocal roots  $\{z_1, \dots, z_p\}$  of their characteristic equations. Assuming for simplicity that both time series are pure AR( $p$ ) processes, they have representations  $y_t^{\text{credit}} = (1 - z_1^{\text{credit}})^{-1} \cdot (1 - z_2^{\text{credit}})^{-1} \cdot \dots \cdot (1 - z_p^{\text{credit}})^{-1} \varepsilon_t^{\text{credit}}$  and  $y_t^{\text{GDP}} = (1 - z_1^{\text{GDP}})^{-1} \cdot (1 - z_2^{\text{GDP}})^{-1} \cdot \dots \cdot (1 - z_p^{\text{GDP}})^{-1} \varepsilon_t^{\text{GDP}}$ . When forming the credit to GDP ratio  $y_t^{\text{credit}}/y_t^{\text{GDP}}$ , common (or similar) roots in numerator and denominator will (almost) cancel out and it is unclear what kind of dynamics the resulting polynomial actually shows. Therefore, the cyclical properties of the credit to GDP ratio are very difficult to interpret.

A clearer result of our univariate analysis is that for none of the countries considered equity shows any of the financial cycle characteristics. This finding is consistent with (near) market efficiency. If equity prices are a good approximation of all the information available to market participants, the strength of serial correlation should be limited.

### 3.3.5. Hypothesis e: The properties of house price cycles and credit cycles are sufficiently similar to assume one financial cycle

A widespread conjecture is that there exists one financial cycle that is common to several financial variables. An important empirical question is, how to combine different variables in order to obtain a single measure of the financial cycle. Drehmann et al. (2012) for instance construct a single time series representing the financial cycle by computing the unweighted average of the medium-term cyclical components of credit, the credit-to-GDP ratio and property prices. The medium-term components are obtained by a Baxter and King (1999) filter that targets frequencies between 8 and 30 years. A similar approach is undertaken by Schuler et al. (2015) who compute a composite financial cycle indicator with time-varying weights of the filtered series of changes in total credit, house prices equity prices and bond yields. English et al. (2005), Hatzius et al. (2010), Ng (2011), Domanski and Ng (2011) and Menden and Proaño (2017) use principal components and dynamic factor models to construct indices of financial market conditions related to the financial cycle. However, they do not explicitly focus on medium-term frequencies. The present paper follows a different approach. Assuming that there might be a common driving force behind different financial variables, we analyze the commonalities of their frequency domain properties.

If the common financial cycle was important, variables which are affected by it should first exhibit similar dynamics and, second, should be related to each other at low frequencies. As to the first aspect, the findings from our previous univariate analysis indicate that similar cyclical properties are only found for credit and house prices, which both show long cycles with large amplitudes. Therefore, we start our multivariate analysis by looking at these two variables within each country.<sup>12</sup>

As to the second aspect, it remains the question where to suspect a low frequency relation; between the levels of credit and house prices, the growth rates or both? Since the levels of credit and house prices are non-stationary, a cointegration relation would represent the interesting case of a common long-run trend, i.e. a relation at the lowest possible frequency  $\lambda = 0$ . However, a probably more popular view in the literature is to think of the common cycle as a low frequency common component of very persistent but still stationary financial variables, like the growth rates of credit and house prices. Evidently, focusing only on the growth rates could lead to the conclusion that no relation exists while in fact there is a relation in levels. The same holds when focusing only on levels.

Therefore, we estimate vector error correction models (VECM) in the time domain to have a representation that separates the relations in levels from the one in first differences and allows us to test for cointegration. After transforming the VECMs into the frequency domain, we consider the coherency measure. A convenient feature of the coherency is that it is invariant to linear transformations and therefore remains unchanged when moving from the levels to first differences. Therefore, we can also use the coherency for the first differences to check at which frequencies we have the strongest correlation between the growth rates of house prices and credit.

Note that the first differences in the VECMs represent quarterly growth rates. This is another interesting aspect, because it allows us to check whether our results from the univariate analysis of yearly growth rates are robust to using quarterly growth rates.

*The relation between credit and house prices within countries.* A formal test of the hypothesis of a common cycle, for example in the series of US credit and house prices, is to check whether there is significant interaction between the levels and the first differences of these variables. However, this would only be a necessary condition. The sufficient condition would be that the interaction takes place at low frequencies. We start with an analysis of the levels. Table 6 shows cointegration test results. In all countries and for both sample periods, we find one significant cointegration relation between credit and house prices.<sup>13</sup> We also obtain clear-cut evidence against a second cointegration relation, which formally implies that the data are in fact non-stationary and contain one common stochastic trend. Another noteworthy result is provided by the adjustment to the equilibrium (not reported in Table 6). In the UK only house prices adjust significantly to equilibrium errors and credit appears to be weakly exogenous. For the second subsample, this also holds for Germany. As will be explained later in more detail, this is in line with the results in Section 4. The frequency domain causality tests in Section 4 suggest that a US-driven global financial cycle affects the other countries mainly through credit. This could explain that credit seems weakly exogenous in

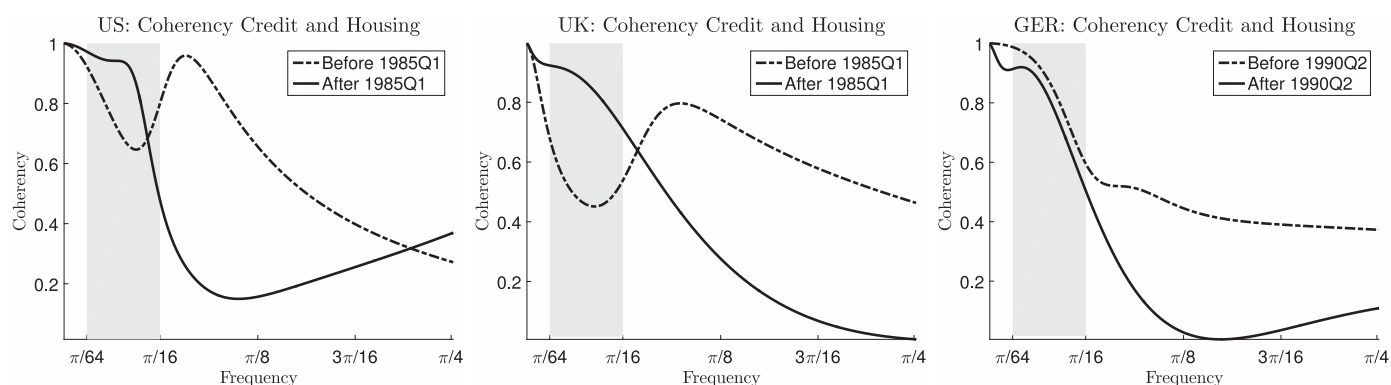
<sup>12</sup> Note that a multivariate analysis is compatible with a univariate study. Even if the true data generating process was a VAR, there still exists a univariate ARMA representation of each time series in the system, see e.g. Maravall and Mathis (1994).

<sup>13</sup> As an anonymous referee pointed out, economically one would also expect a cointegration relation between the variables credit volume and real total value of the housing stock - as opposed to house prices.

**Table 6**  
Cointegration tests - Credit and house prices.

|     | Pre          |      |              |                        | Post         |      |              |                        |
|-----|--------------|------|--------------|------------------------|--------------|------|--------------|------------------------|
|     | $H_0$ (rank) | Lags | Trace test   | $\lambda_{\max}$ -test | $H_0$ (rank) | Lags | Trace test   | $\lambda_{\max}$ -test |
| US  | $r = 0$      | 4    | 35.4 (0.002) | 29.1 (0.001)           | $r = 0$      | 5    | 28.0 (0.027) | 21.1 (0.028)           |
|     | $r \leq 1$   | 4    | 6.3 (0.419)  | 6.3 (0.419)            | $r \leq 1$   | 5    | 6.9 (0.355)  | 6.9 (0.355)            |
| UK  | $r = 0$      | 3    | 24.1 (0.083) | 17.0 (0.106)           | $r = 0$      | 2    | 23.3 (0.102) | 18.0 (0.078)           |
|     | $r \leq 1$   | 3    | 7.0 (0.343)  | 7.0 (0.343)            | $r \leq 1$   | 2    | 5.2 (0.562)  | 5.2 (0.562)            |
| GER | $r = 0$      | 4    | 36.3 (0.002) | 27.2 (0.003)           | $r = 0$      | 4    | 26.0 (0.048) | 20.2 (0.038)           |
|     | $r \leq 1$   | 4    | 9.0 (0.179)  | 9.0 (0.179)            | $r \leq 1$   | 4    | 5.8 (0.483)  | 5.8 (0.483)            |

Notes: This table shows the Johansen (1995) cointegration test results. We use two test statistics, the  $\lambda_{\max}$  and the trace statistic. The lag length of each VECM is specified by using standard information criteria and residual autocorrelation tests to confirm that the lag order is sufficient to guaranty white noise residuals. Since all series are trending, the deterministic terms in the VECMs allow for different drifts of the time series. Pre and post refer to the sample periods before and after the break.



**Fig. 4.** Coherency Between Credit and House Prices. Note: This Figure shows the coherency between credit and house prices. Since the coherency is invariant to linear transformations, it applies to the levels and to the first differences of the time series.

the present analysis of credit and housing within the countries UK and Germany.

The long-run equilibrium relation of the variables is clearly visible in the frequency domain when regarding the coherencies in Fig. 4. As implied by cointegration, at frequency  $\lambda = 0$  the coherency is equal to 1. Fig. 4 is also very informative about the interaction at cycles of finite length. It can be seen that for the US and UK the center of correlation has shifted from business cycle frequencies before the sample split to much lower frequencies thereafter. For the case of Germany, the coherency remains more stable.

Results for the dynamics of each quarterly growth rate are provided in Fig. 5. We derived the individual spectral densities using the estimated VECM parameters as outlined in Section 2.3. While the shapes of the spectral densities differ to some extent, the overall picture is fairly in line with the one obtained from the univariate analysis in the previous section.

As explained in detail in B.2, this can be interpreted as follows. If most of the spectral mass of the true data generating process is located at frequencies lower than, say,  $\pi/16$ , it makes only little difference between using the  $(1-L)$ -filter and the  $(1-L^4)$ -filter. Acting similar at low frequencies, both filters will provide an accurate picture of the cyclical properties of the stationary component of the times series under investigation.

*The relation between financial variables and GDP within countries.* To provide some indicative evidence on how the financial cycle may affect real economic activity we also calculated the coherencies between GDP and credit as well as GDP and house prices. The results are shown in Fig. 6. The interaction between the financial variables and GDP is more pronounced at lower frequencies in the second

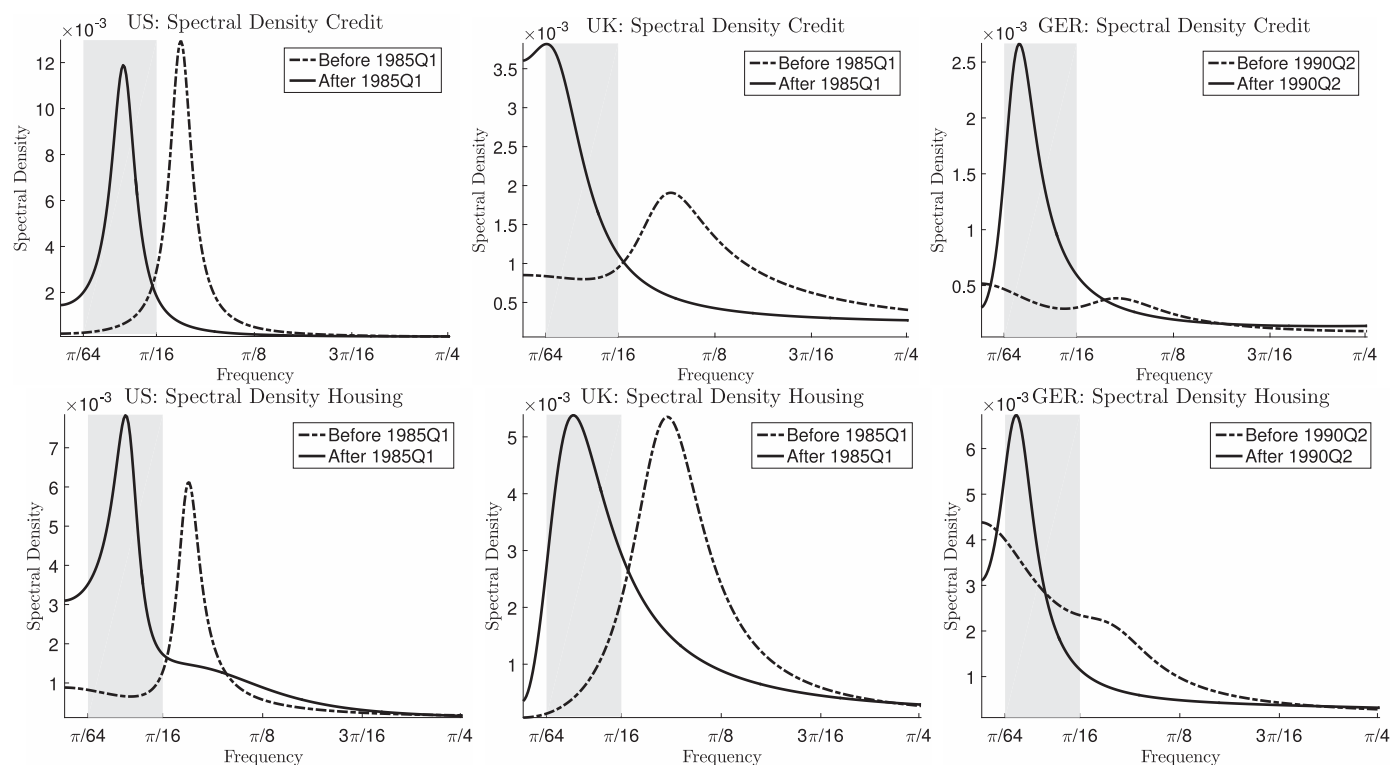
subsample compared to the first one. The coherencies somewhat suggest a cointegration relation in a few cases but the evidence is not fully convincing.<sup>14</sup> In any case, it is valid to interpret the coherencies as describing the relation between the quarterly growth rates of the time series.

*The relation between financial variables across countries.* From the above multivariate analysis of credit and house prices within countries, we have found correlation at very low frequencies. However, this can only be a first indication of a common cycle. In order to go a little further towards a more detailed analysis of the global cycle, we estimate four bivariate cross-country VARs consisting of the US plus either the UK or Germany. We now look at frequency-wise Granger-causality for credit and house prices. The Granger-causality across frequencies is calculated according to (10) and measures the causality from the US to the second country in the system. Fig. 7 shows the results. While the exact magnitude of the measures is not informative, it can be observed that the impact of the US on the UK and on Germany is concentrated around specific frequencies and has moved over time towards lower ones.

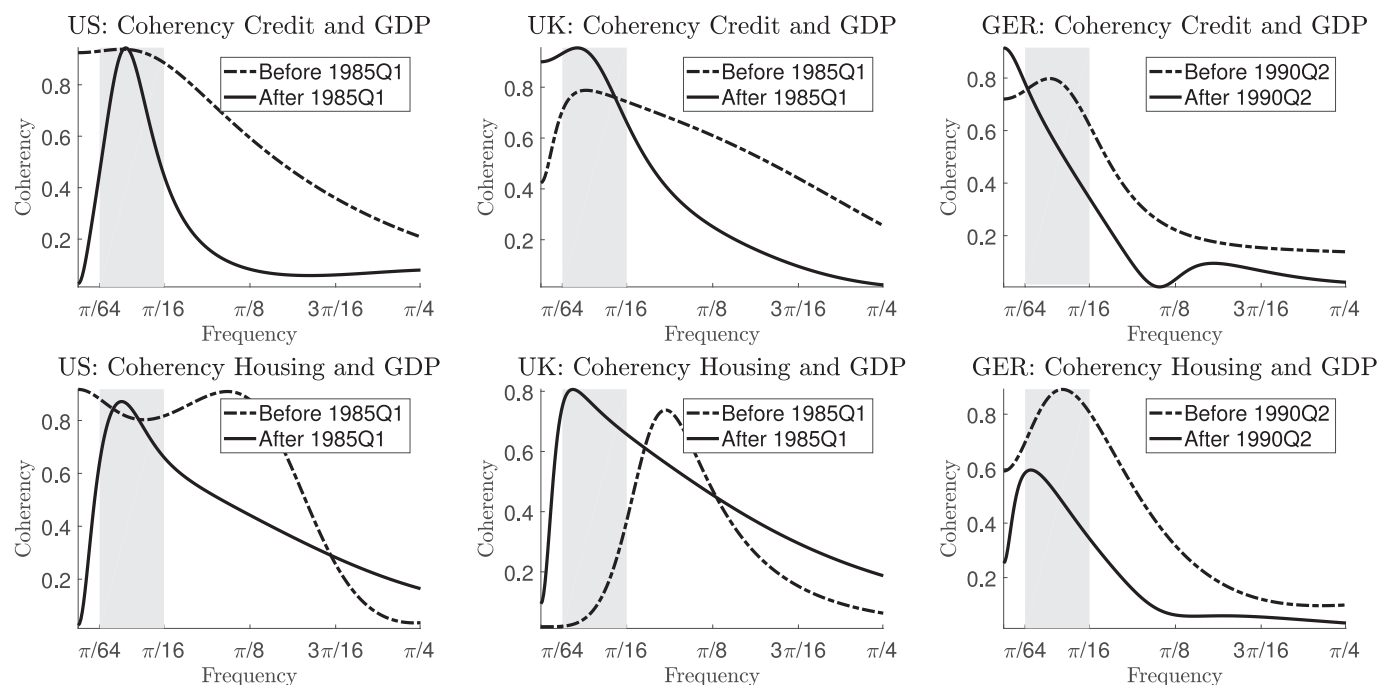
Further indicative evidence for a global financial cycle is obtained by an investigation of global credit, a series provided by the Bank for International Settlements. We performed cointegration tests of global credit and the national credit series as well as tests for weak exogeneity. The results are summarized in Table 7.<sup>15</sup> Two findings stand out. First, evidence for cointegration between

<sup>14</sup> This is also confirmed by time domain cointegration test results which allow only for a few cases for a (quite weak) rejection of the null of no cointegration.

<sup>15</sup> Due to data availability, the analysis relies on a limited sample from 1999Q1 to 2013Q4. In line with the national credit series, global credit measures credit to the



**Fig. 5.** Spectral Densities of First Differences. *Note:* Individual spectral densities for quarterly growth rates of credit and house prices derived from country-specific VECM representations.

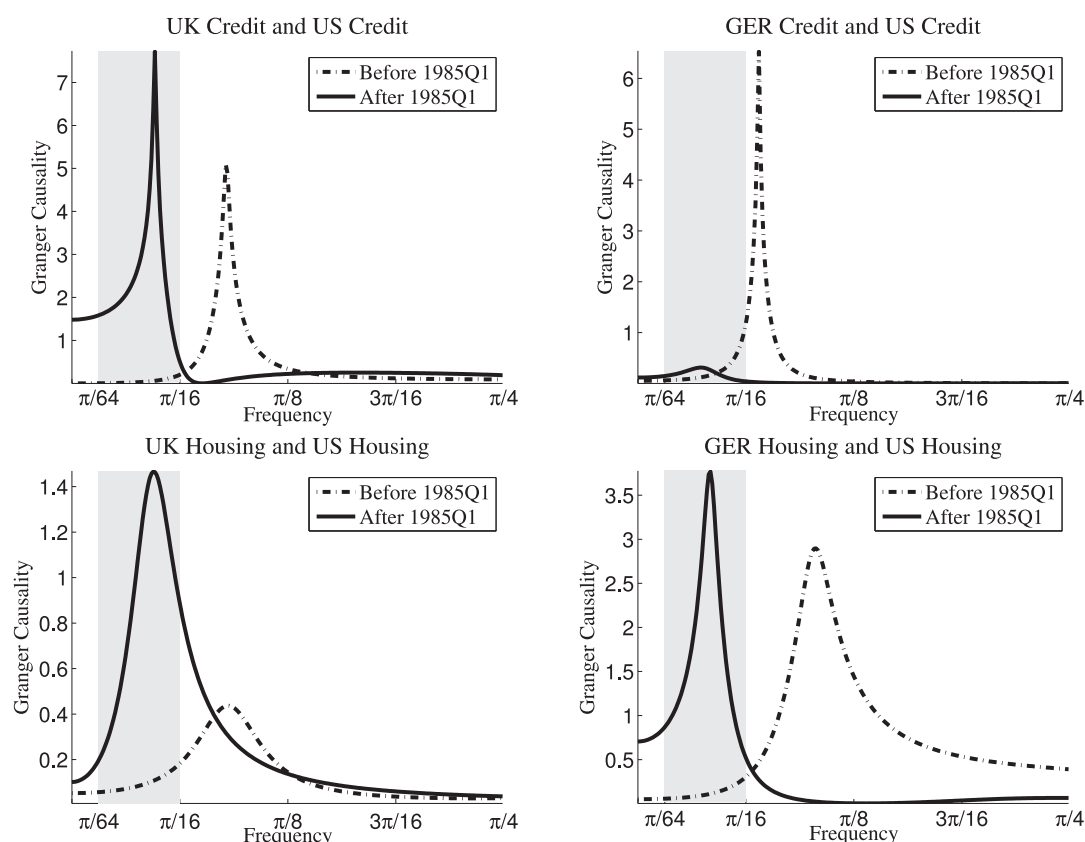


**Fig. 6.** Coherency Between GDP and Credit (first row) and GDP and House Prices (second row). *Note:* This Figure shows the coherencies between GDP and credit and GDP and house prices. Since the coherency is invariant to linear transformations, it applies to the levels and to the first differences of the time series.

national credit and global credit is quite clear for all three countries. Second, for UK and Germany, global credit is not signifi-

cantly adjusting to deviations from the long-run relation and is thus weakly exogenous. In contrast, for the US, global credit adjusts significantly to equilibrium errors while US credit does not. The weak exogeneity property of US credit provides some first evidence that a global financial cycle is likely to be driven by the US.

private non-financial sector from all sectors. The data are obtained from the Bank for International Settlements data base.



**Fig. 7.** Cross-Country Granger-Causality *Note:* This Figure shows the Granger-causality measure of [Breitung and Candelon \(2006\)](#). The measures are derived from bivariate cross-country VARs and show the causality from the US to the second country in the system.

**Table 7**  
Cointegration tests - Global credit and national credit.

|     | 1999Q1 – 2013Q4 |      |              |                        | Weak exogeneity |               |
|-----|-----------------|------|--------------|------------------------|-----------------|---------------|
|     | $H_0$ (rank)    | Lags | Trace test   | $\lambda_{\max}$ -test | National credit | Global credit |
| US  | $r = 0$         | 4    | 26.0 (0.048) | 16.8 (0.116)           | yes             | no            |
|     | $r \leq 1$      | 4    | 9.3 (0.164)  | 9.3 (0.164)            |                 |               |
| UK  | $r = 0$         | 1    | 35.3 (0.003) | 26.9 (0.003)           | no              | yes           |
|     | $r \leq 1$      | 1    | 8.4 (0.219)  | 7.0 (0.219)            |                 |               |
| GER | $r = 0$         | 1    | 29.2 (0.018) | 25.0 (0.007)           | no              | yes           |
|     | $r \leq 1$      | 1    | 4.2 (0.708)  | 9.0 (0.708)            |                 |               |

*Notes:* This table shows the [Johansen \(1995\)](#) cointegration test results for national credit series and global credit. Weak exogeneity likelihood ratio tests are performed by setting the adjustment coefficient equal to zero.

#### 4. Does US monetary policy affect credit and housing cycles overseas?

The significant comovement of bond and equity prices in many economies around the world has been explained by [Rey \(2013\)](#) through the existence of a global financial cycle related to changes in general risk appetite and market uncertainty.<sup>16</sup> [Rey \(2013\)](#), [Miranda-Agrippino and Rey \(2015\)](#) and [Passari and Rey \(2015\)](#) also show that the global financial cycle is highly correlated with the VIX volatility index of the Chicago Stock Exchange

and that the VIX, in turn, is influenced by US monetary policy (see also [Borio and Zhu, 2012](#), [Bekaert et al., 2013](#) and [Bruno and Shin, 2015](#)). Hence, the global financial cycle represents an important additional transmission channel of US monetary policy at the international level primarily through the bond and equity markets. Moreover, as the global cycle cannot be aligned with the macroeconomic conditions of all countries at the same time, it represents a particular challenge for the conduct of national monetary policies.

By contrast, we investigate in this section the possible linkage between the US VIX and the national credit and housing cycles. The frequency domain analysis in the previous section provided some indicative evidence for a significant comovement in the growth rates of credit and house prices. As a natural next step we now extend the bivariate VARs by a number of key macroeco-

<sup>16</sup> The notion of the existence of such a common financial cycle is not new: Seminal work by [Fisher \(1933\)](#), [Keynes \(1936\)](#), [von Mises \(1952\)](#), [Hayek \(1933\)](#) and [Minsky \(1982\)](#) stressed the inherently procyclical behavior of the financial system and the role of bounded rational behavior by financial market participants.

**Table 8**Time domain Granger non-causality tests (*p*-values), Estimation sample: 1990:1–2013Q4.

| ...is not Granger-caused by... | Credit | GDP   | RER   | Housing prices | Real interest rate | VIX   |
|--------------------------------|--------|-------|-------|----------------|--------------------|-------|
| United States                  |        |       |       |                |                    |       |
| Credit                         | –      | 0.231 | –     | 0.001          | 0.640              | 0.363 |
| GDP                            | 0.211  | –     | –     | 0.154          | 0.257              | 0.370 |
| RER                            | –      | –     | –     | –              | –                  | –     |
| House Prices                   | 0.009  | 0.093 | –     | –              | 0.132              | 0.175 |
| Real Interest Rate             | 0.771  | 0.984 | –     | 0.948          | –                  | 0.638 |
| VIX                            | 0.853  | 0.172 | –     | 0.122          | 0.520              | –     |
| United Kingdom                 |        |       |       |                |                    |       |
| Credit                         | –      | 0.205 | 0.814 | 0.445          | 0.553              | 0.094 |
| GDP                            | 0.359  | –     | 0.509 | 0.120          | 0.979              | 0.001 |
| RER                            | 0.906  | 0.364 | –     | 0.004          | 0.124              | 0.265 |
| House Prices                   | 0.059  | 0.853 | 0.106 | –              | 0.999              | 0.304 |
| Real Interest Rate             | 0.046  | 0.439 | 0.029 | 0.517          | –                  | 0.274 |
| VIX                            | 0.714  | 0.104 | 0.545 | 0.641          | 0.181              | –     |
| Germany                        |        |       |       |                |                    |       |
| Credit                         | –      | 0.001 | 0.318 | 0.003          | 0.698              | 0.509 |
| GDP                            | 0.171  | –     | 0.118 | 0.458          | 0.338              | 0.376 |
| RER                            | 0.545  | 0.191 | –     | 0.421          | 0.879              | 0.872 |
| House Prices                   | 0.778  | 0.495 | 0.622 | –              | 0.242              | 0.232 |
| Real Interest Rate             | 0.521  | 0.184 | 0.596 | 0.947          | –                  | 0.639 |
| VIX                            | 0.474  | 0.100 | 0.709 | 0.257          | 0.319              | –     |

Note: The direction of the Granger-causality presented in this table goes from columns to rows, i.e. column variables Granger-cause row variables.

economic variables in order to investigate in more detail if and how the global financial cycle affects national credit and housing cycles from a frequency domain perspective.

The lag order of the following VAR models was determined through the Schwarz Information Criterion and, if necessary, adjusted until all autocorrelation was removed from the residuals. Due to data availability of the VIX series, which starts only at 1990Q1, the estimation sample for the following VAR analysis is 1990Q1–2013Q4. For the US, we estimate a VAR model consisting of the variables credit, real GDP, house prices (all three in logs), the real short-term interest rate (calculated as in Bruno and Shin, 2015 as the target Fed Funds rate minus the CPI inflation rate) and the log VIX, in this order. The models for the UK and Germany include the real exchange rate as an additional endogenous variable. This reflects that the US may be considered a large and rather closed economy while the UK and Germany are comparatively small and open economies. Besides, the exchange rate will control for the traditional transmission mechanism of US monetary policy on other countries.

A similar approach is followed by Rey (2016). However, her analysis is conducted solely in the time domain. In our study, to obtain further insights into the international transmission of US monetary policy, we additionally analyze the influence of the global financial cycle on national credit and housing cycles in the frequency domain using the Breitung and Candelon (2006) test.<sup>17</sup> As the time domain counterpart, we apply the Toda and Yamamoto (1995) procedure.<sup>18</sup> For both methods the null hypothesis is no Granger-causality.

Let us first investigate whether the relation between house price growth and credit growth in the US found in the previous sections can be reproduced in the larger and more com-

plete VAR model. And, are these variables influenced by the US VIX?

In Table 8 we report statistical evidence supporting the hypothesis that US house prices are Granger-causal and thus an important driver of the dynamics of US credit (*p*-value = 0.001). In addition to the time domain causality test, the first graph in the first row of Fig. 8 illustrates that the Granger-causality of US house prices on US credit occurred primarily on medium-term frequencies. Looking into the opposite direction, US house prices can also be considered as being Granger-caused by US credit at standard significance levels. However, as the second graph in the first row of Fig. 8 shows, the influence from credit to prices is statistically significant in the whole frequency range considered. These findings are in line with the widespread notion that the self-reinforcing interaction between house prices and credit was a key factor of the 2007/2008 crisis and the following collapse in economic growth (see e.g. Geanakoplos, 2010). Further, while US GDP does not appear to be significantly influenced by housing or credit in the time domain – at least not at the 10%-level – looking in the frequency domain provides a more detailed picture. From the third graph in the first row of Fig. 8 it can be seen that house prices do significantly affect GDP, but only at lower frequencies. Turning to the US VIX, Table 8 suggests that there is no Granger-causality on any of the other variables in the US VAR. However, Fig. 8 (second graph in the first row) reveals significant low frequency Granger-causality from the VIX to house prices.

We now address the following question: Does US monetary policy transmit through the risk-taking channel and affect national credit and housing cycles in other countries? If so, there should be significant Granger-causality at low frequencies between the US VIX and foreign credit (or house prices).

According to the *p*-values for the time-domain Granger-causality tests reported on Table 8, the US volatility index can be considered as the sole Granger-causal variable for both credit and real GDP in the UK. In the frequency domain, we see that the significance of the Granger-causality relationships stems from low and medium-term frequencies (first and third graph in the second row of Fig. 8). The same result holds for the effect of the US VIX on macroeconomic activity in the UK. These findings support the hypothesis of a global financial cycle, associated with the US VIX, which affects national credit and house price cycles. However, our

<sup>17</sup> As in the previous section, the spectral matrix of the VAR models is derived from the estimated time domain parameters. Intuitively, one can think of the *p*-values of time-domain Granger-causality tests as the summary or average of the frequency domain Granger-causality over the whole spectrum. Therefore, it is possible that time domain Granger-causality may be insignificant at standard confidence levels, while for certain frequencies the relation is in fact significant.

<sup>18</sup> As our VARs contain *I*(1) and *I*(0) variables, a standard Granger-causality test cannot be applied as the Wald test statistic does not follow its usual asymptotic  $\chi^2$  distribution under the null.

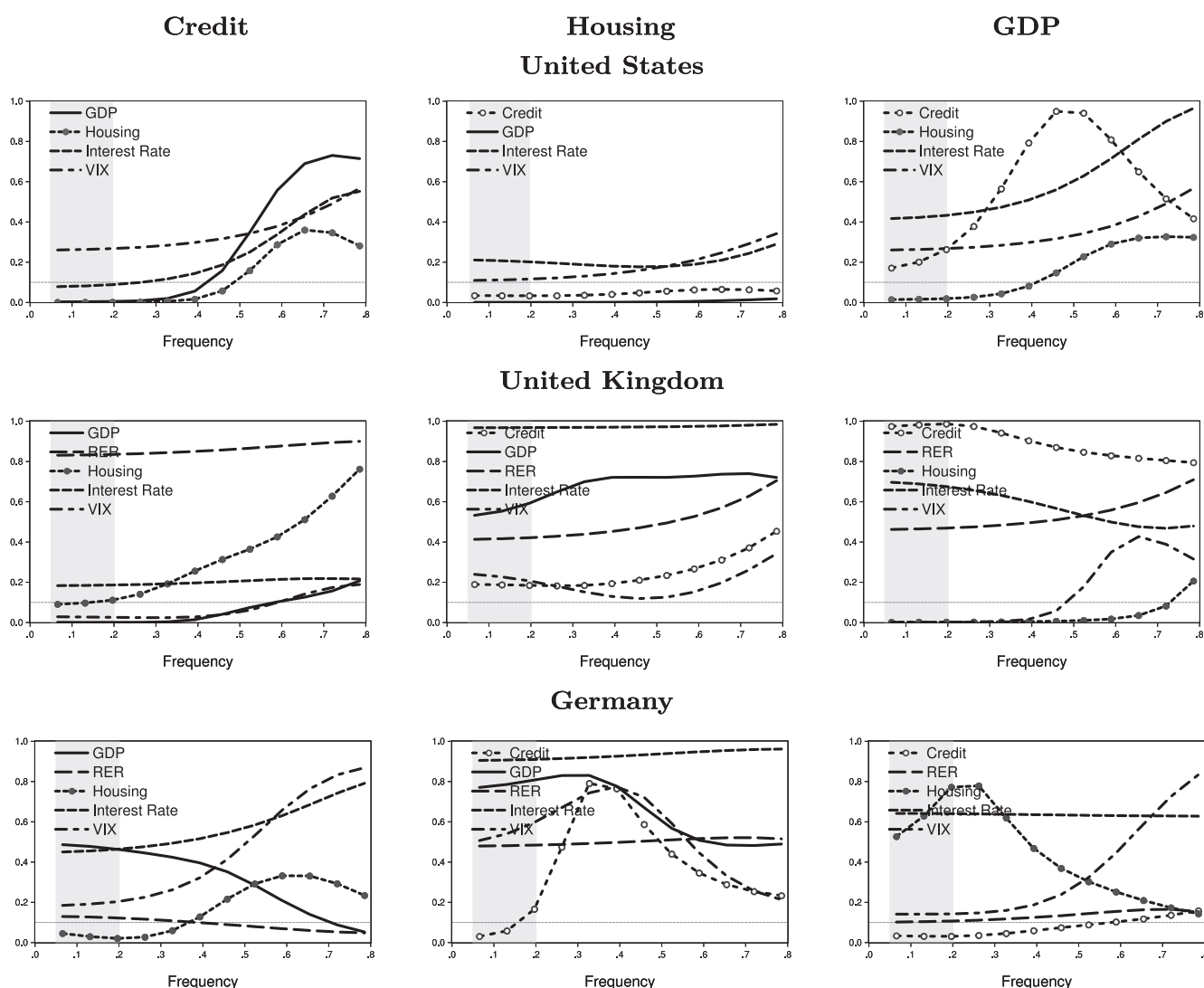


Fig. 8. Breitung and Candelon (2006)  $p$ -values for frequency domain Granger non-causality (low  $p$ -values indicate a rejection of the null hypothesis of non-causality). Shaded areas represent medium-term frequencies.

study shows that for a comprehensive understanding of the relations is it essential to consider both the time and the frequency domain perspective.

For Germany, we obtain less clear-cut results. The time domain tests show no significant Granger-causality from the VIX to German financial variables and to real economic activity. However, the frequency domain results in Fig. 8 are again very informative. The first and the third graph in the last row show that the significance of Granger-causality from the VIX to credit and GDP clearly increases towards lower frequencies, approaching levels of less than 20%. Therefore, it could well be the case that the higher dimensional VAR for Germany would indicate similar results as for the UK after some years, if the German economy becomes as sensitive to global financial factors as the UK. Yet, in the present sample, no significant effects are visible.

## 5. Concluding remarks

What are the main characteristics of the financial cycle? Is it a medium-term phenomenon, meaning it is longer than the busi-

ness cycle, as suggested in the literature, or does it share similar characteristics with the business cycle? How are financial cycles related within countries and what could explain internationally related financial cycles? In this paper we intended to shed some light on these and other related questions by estimating the data generating processes of financial and business cycle variables using standard econometric methods. We extended the literature by formally testing various cyclical properties and economic interpretations that have been suggested by existing work. Most of our analysis of the financial cycle was performed in the frequency domain, while the estimation was carried out in the time domain. Our approach allowed us to take into account all possible cycles without *a priori* assuming different ranges for financial and business cycles. The efficiency of the parametric spectrum estimation made it possible to test for time-variation of cyclical properties via subsample analysis and to test for Granger-causality in a multivariate setting.

For the US and the UK, we found strong evidence that growth rates of credit and house prices feature longer cycles and larger amplitudes than the business cycle. We also found time variation

of these properties and showed that they have become more visible in recent times. The statistical significance of our findings contributes to establishing stylized facts of the properties of financial cycles. Further, we found statistical evidence at low frequencies supporting the relevance of Rey's (2013) global financial cycle hypothesis and thus of the risk-taking channel of US monetary policy on the credit and housing cycles of the UK in particular.

Our results highlight the need for closely monitoring the evolution of financial cycles and for policy-measures to curb extreme fluctuations in order to mitigate the effects of the global financial cycle.

An interesting and straightforward extension of our approach that could be applied in both a univariate and a multivariate framework would be to use time-varying parameter models. These could possibly provide deeper insights into the question of *when* the financial cycle has developed its distinguishing features and *what* events have caused the parameter changes.

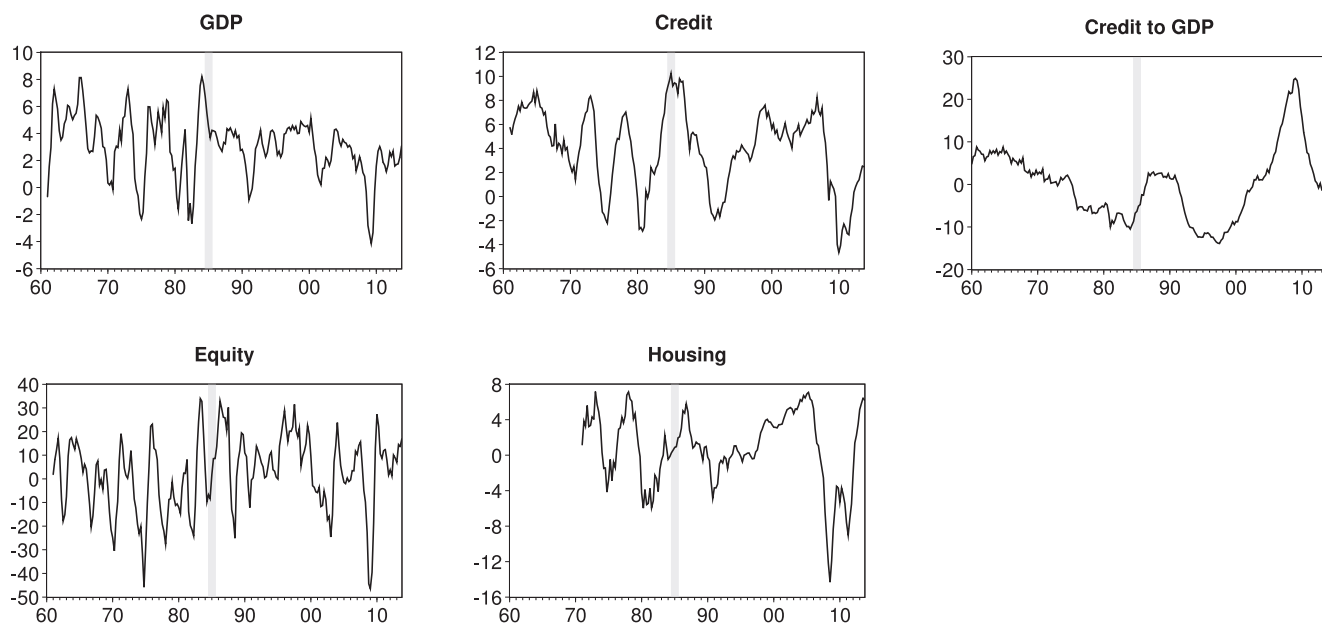
### Acknowledgements

We are grateful for comments and suggestions received from the editor Geert Bekaert and two anonymous referees. We also thank Helmut Lütkepohl, Dieter Nautz, Christian Offermanns, Sven Schreiber, Lars Winkelmann, an anonymous referee from the Bundesbank Discussion Paper series, as well as participants of the Eco-

nomics Seminar at Otago University, Dunedin, New Zealand, the Economics Seminar at the University of Canterbury, Christchurch, New Zealand, the FU Berlin Macroeconomic Workshop, the FU Berlin Research Seminar in Economics, the 35<sup>th</sup> International Symposium on Forecasting, Riverside, California, the 2<sup>nd</sup> Annual Conference of the International Association for Applied Econometrics, Thessaloniki, Greece, the 15<sup>th</sup> INFINITI Conference on International Finance, Trinity College Dublin, Ireland and the Swiss National Bank Research Seminar for helpful comments on an earlier version of this paper. The research was partly conducted while Christian Proaño was a visiting researcher at the Bundesbank, whose hospitality is gratefully acknowledged. Financial support from the Deutsche Forschungsgemeinschaft (DFG) through CRC 649 “Economic Risk”, as well as by the Hans-Böckler Foundation is also gratefully acknowledged.

### Appendix A. Data sources and definitions

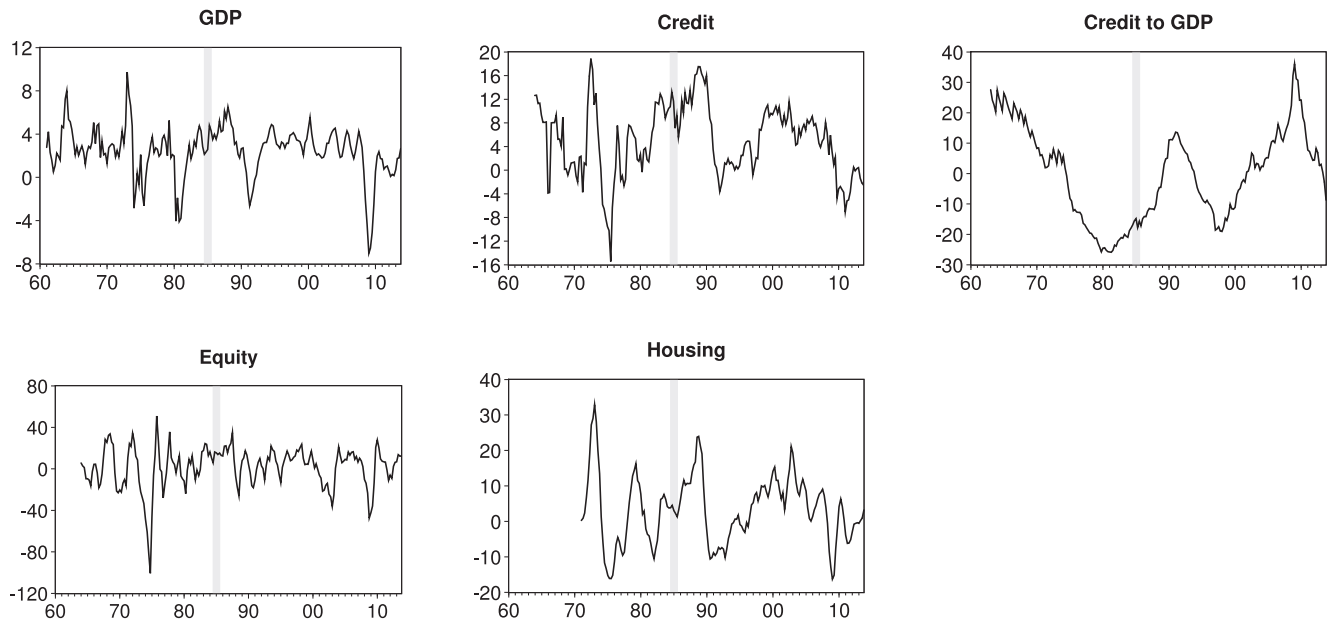
All series are measured in logs and deflated using the consumer price index. All series are normalized by their respective value in 1985Q1 to ensure comparability of the units. We obtain annual growth rates by taking annual differences of the time series. The only exception is the credit to GDP ratio which is expressed in percentage points (Table 9).



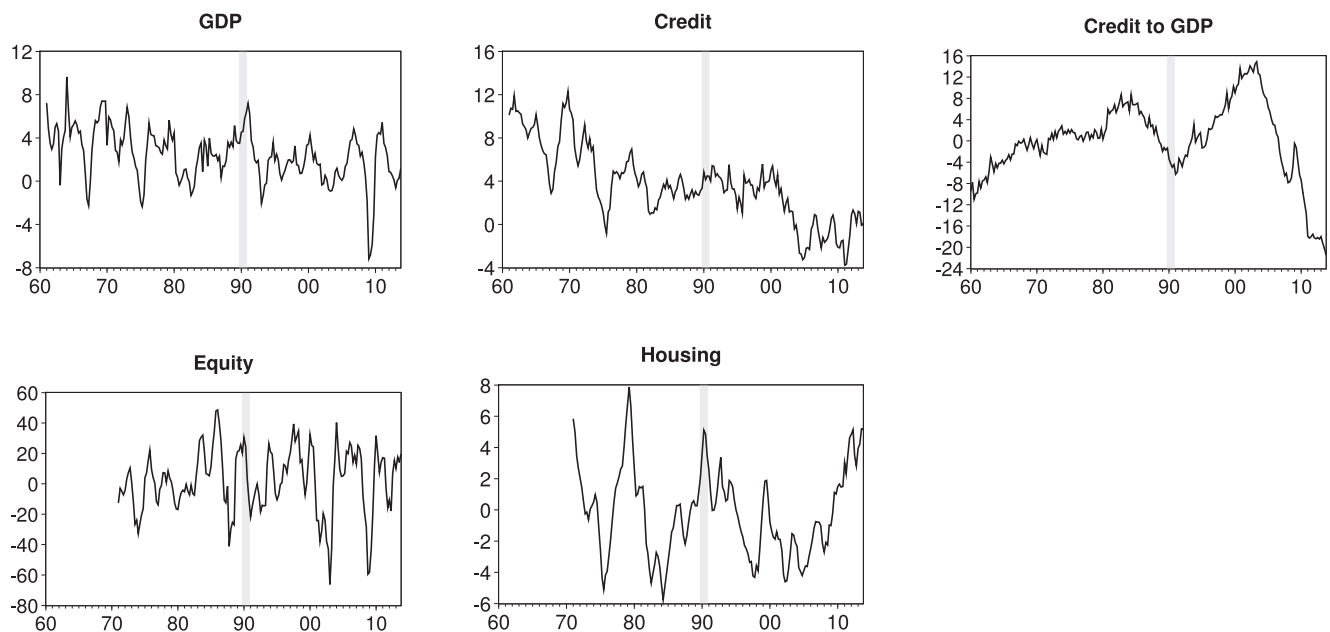
**Fig. 9.** Real GDP and Financial Variables in the United States. *Note:* All series are annual growth rates, except the credit to GDP ratio, which represents deviations from a linear trend measured in percentage points. The vertical gray line shows the sample split.

**Table 9**  
Definition and sources of the data.

|         | Source     | Identifier                       | Notes                                                                      |
|---------|------------|----------------------------------|----------------------------------------------------------------------------|
| GDP     | OECD.Stat  | CARSA                            | National currency                                                          |
| CPI     | OECD.Stat  | Consumer Prices                  | National index                                                             |
| Credit  | Datastream | USBLCAPAA, UKBLCAPAA, BDBLCAPAA  | National currency, credit to private Non-financial sector from all sectors |
| Housing | OECD.Stat  | House Prices                     | National index                                                             |
| Equity  | IMF        | USQ62... F, UK62... F, BDQ62.EPC | National index                                                             |



**Fig. 10.** Real GDP and Financial Variables in the United Kingdom. *Note:* All series are annual growth rates, except the credit to GDP ratio, which represents deviations from a linear trend measured in percentage points. The vertical gray line shows the sample split.



**Fig. 11.** Real GDP and Financial Variables in Germany. *Note:* All series are annual growth rates, except the credit to GDP ratio, which represents deviations from a linear trend measured in percentage points. The vertical gray line shows the sample split.

## Appendix B. Methodological discussion

### B.1. Comparison of indirect spectrum estimation and filtering

A popular approach in the analysis of the financial cycle is to work with frequency-based filters (Drehmann et al., 2012; Aikman et al., 2015) which are usually based on Baxter and King (1999) or refinements thereof, see e.g. Christiano and Fitzgerald (2003). When filters are used to analyze cycles, a crucial point is that the frequency range has to be pre-specified by the researcher. This may

lead to results biased into the direction of the pre-defined range. The intuition behind this difficulty is best described by briefly discussing the general functioning of filters.

In the time domain, any linear filter can be written in the form of a two-sided moving average

$$y_t = \sum_{j=-m}^n a_j x_{t-j}. \quad (13)$$

The filtered series  $y_t$  depends simultaneously on the properties of the filter (the coefficients  $a_j$ ) and the data (the observed series  $x_t$ ).

To convert Eq. (13) into the frequency domain, one uses the corresponding filter function

$$C(\lambda) = \sum_{j=-m}^n a_j e^{-i\lambda j}, \quad i^2 = -1, \quad (14)$$

and its transfer function

$$T(\lambda) = |C(\lambda)|^2, \quad (15)$$

where  $\lambda \in [-\pi, \pi]$  denotes the frequency range, see e.g. Wolters (1980a,b). The relationship between the spectra of the original and the filtered series is given by

$$f_y(\lambda) = T(\lambda) f_x(\lambda). \quad (16)$$

When compared to the time domain, a major advantage of the representation in (16) is that now the filter effect  $T(\lambda)$  is separated from the data  $f_x(\lambda)$ .

For the spectrum of the filtered series  $f_y(\lambda)$  to accurately identify cycles in the spectrum of the observed series  $f_x(\lambda)$ , a so-called band pass filter must be used. Such a filter has the value 1 at the frequency range of interest  $[\lambda_{\text{low}}, \lambda_{\text{high}}]$  and 0 otherwise. In view of Eq. (16), this implies  $f_y(\lambda) = f_x(\lambda)$  for  $\lambda \in [\lambda_{\text{low}}, \lambda_{\text{high}}]$  and  $f_y(\lambda) = 0$  for  $\lambda \notin [\lambda_{\text{low}}, \lambda_{\text{high}}]$ . Theoretically, the ideal time domain filter can be achieved by moving averages of infinite order. Yet, in practice, this is not possible since only a limited number of observations is available.

The main idea of frequency-based filters is to pre-specify a range from  $\lambda_{\text{low}}$  to  $\lambda_{\text{high}}$  and to choose finite values for  $m$  and  $n$  in Eq. (13) to find the weights  $a_j$  which approximate the ideal filter as good as possible. Due to the approximation, the spectrum of the filtered series is in general different from the one of the observed series in  $[\lambda_{\text{low}}, \lambda_{\text{high}}]$  and reflects a mixture of filter and data properties in the form of  $T(\lambda)f_x(\lambda)$ . More precisely, it can be shown that any filter has the tendency to overstate the contributions of cycles in the pre-defined frequency range  $[\lambda_{\text{low}}, \lambda_{\text{high}}]$  to the overall variance of the underlying time series. Because of this tendency, artificial cycles may be produced even if the true data generating process has no cycles. This issue has been raised already in the 1950/s and 1960/s. For a discussion of the general problem see e.g. König and Wolters (1972), King and Rebelo (1993), Baxter and

King (1999), Christiano and Fitzgerald (2003), Harvey and Trimbur (2003) and Murray (2003).

In this study, we emphasize that the appropriate  $\lambda_{\text{low}}$  and  $\lambda_{\text{high}}$  are unknown. Indeed, while there seems to be general consensus in the literature about the relevant frequency range for the business cycle (2 to 8 years), the relevant frequency range for the financial cycle is less clear. For instance, Drehmann et al. (2012) construct their financial cycle measure from the underlying series with a *a priori* chosen cycle length between 8 and 30 years.

## B.2. The effect of quarterly and yearly growth rate filters: $\Delta$ vs. $\Delta_4$

Our univariate analysis of financial cycles focuses on yearly growth rates of a number of financial variables. This allows us to compare our results to those provided by existing studies. The fact that most of the literature focuses on yearly growth rates also highlights that the dynamics of yearly changes are of general interest to economists.

Nonetheless, when calculating yearly growth rates, a specific filter, the  $(1 - L^4)$ -filter, is applied to the level of the time series. This raises the concern if and how this filter changes the cyclical properties of the data. Another interesting aspect is whether it makes a difference when using the  $(1 - L)$ -filter to calculate quarterly instead of yearly growth rates.

What we do know about the level data is that they are non-stationary. Hence, the spectral density collapses at frequency  $\lambda = 0$  because all the variation of the series is explained by the cycle with length infinity. Therefore, the more relevant question must be whether using the  $(1 - L^4)$  or  $(1 - L)$ -filter gives a good approximation of the cyclical properties of the stationary component of the time series which remains when the unit root is removed. The correct answer, in turn, depends crucially on the DGP.

Consider the DGP

$$(1 - L^4)A(L)y_t = \varepsilon_t \quad (17)$$

with  $A(L)$  stationary and  $\varepsilon_t$  white noise. The process in (17) is still fairly general and could well be the generating process behind the present data. The true spectrum of the level of  $y_t$  is given by

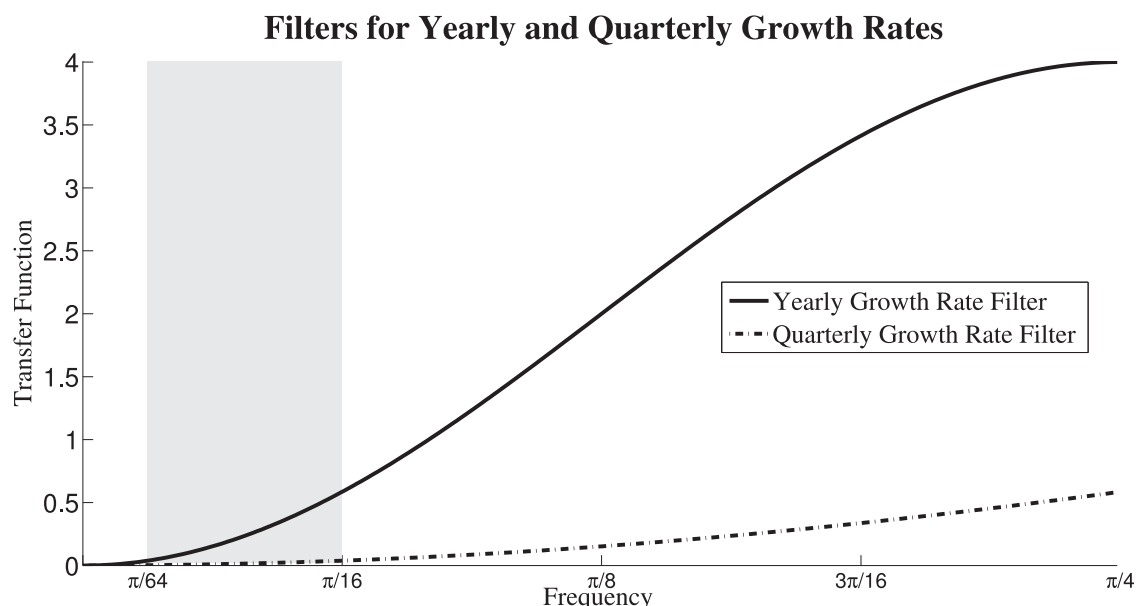
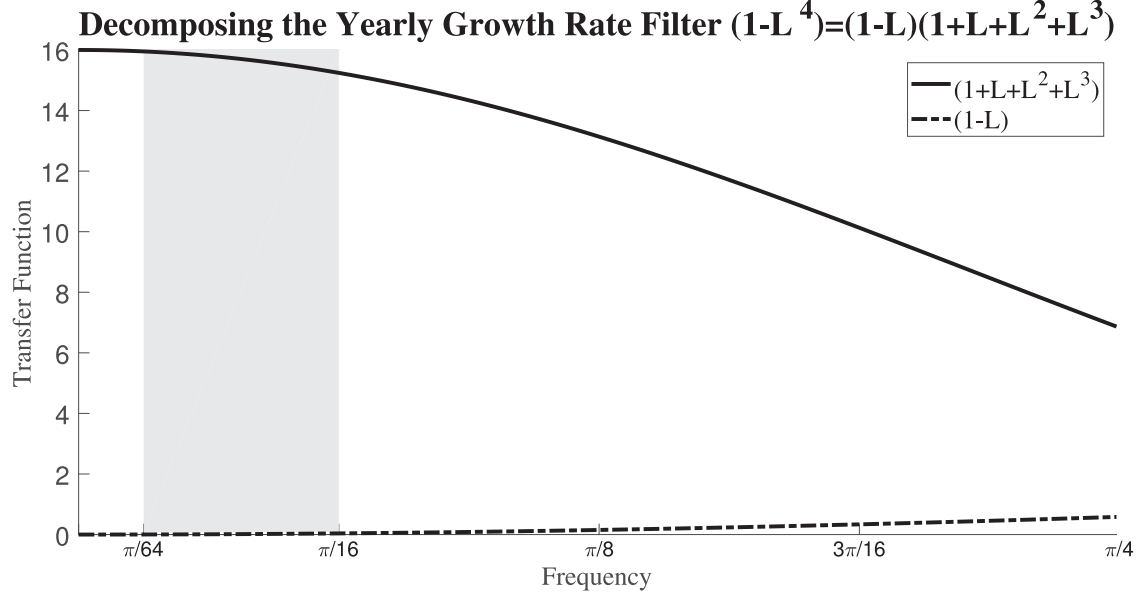


Fig. 12. Comparing the Filters  $(1 - L^4)$  and  $(1 - L)$ . Note: The Figure shows the transfer functions of the yearly and quarterly growth rate filters, i.e.  $2(1 - \cos(4\lambda))$  and  $2(1 - \cos(\lambda))$ .

**Table 10**Values of the transfer function of the  $(1 - L^4)$  and  $(1 - L)$ -filters for medium-term frequencies.

| Frequency   | $\pi/64$ (32 years) | $\pi/48$ (24 years) | $\pi/32$ (16 years) | $\pi/24$ (12 years) | $\pi/16$ (8 years) |
|-------------|---------------------|---------------------|---------------------|---------------------|--------------------|
| $(1 - L)$   | 0.0024              | 0.0043              | 0.0096              | 0.0171              | 0.0384             |
| $(1 - L^4)$ | 0.0384              | 0.0681              | 0.1522              | 0.2679              | 0.5858             |

Notes: This table shows values of the transfer functions for the  $(1 - L)$  and the  $(1 - L^4)$ -filters for selected medium term frequencies.



**Fig. 13.** Decomposing the Filter for Yearly Growth Rates. Note: The figure shows the transfer functions of the two components  $(1 - L)$  and  $(1 + L + L^2 + L^3)$  of the  $(1 - L^4)$ -filter.

$$f_y(\lambda) = [2(1 - \cos(4\lambda))]^{-1} \cdot [|A(e^{-i\lambda})|^2]^{-1} \cdot f_{\varepsilon_t}(\lambda) \\ = [2(1 - \cos(4\lambda))]^{-1} \cdot f_{\Delta_4}(\lambda) \cdot \frac{\sigma_{\varepsilon_t}^2}{2\pi}$$

In that case, the  $(1 - L^4)$ -filter, which has the frequency domain representation  $[2(1 - \cos(4\lambda))]$ , should indeed be applied because it would give us exactly (up to a constant scale factor  $\frac{\sigma_{\varepsilon_t}^2}{2\pi}$ ) the true spectrum of the cyclical component. Of course, the crucial assumption is that true spectrum of the cyclical component is  $f_{\Delta_4}(\lambda) = [|A(e^{-i\lambda})|^2]^{-1}$ . Naturally, the same argument could be made against the  $(1 - L^4)$ -filter by assuming the DGP was  $(1 - L)A(L)y_t = \varepsilon_t$ .

Of course, the true DGP will always remain unknown. In this paper we were interested in cycles in growth rates. In an analysis of cycles in level data, both the  $(1 - L)$ -filter and the  $(1 - L^4)$ -filter will, in general, dampen medium-term frequencies. This can be seen from the transfer functions in Fig. 12 and Table 10. The  $(1 - L)$ -filter would dampen cycles for  $\lambda < \pi/4$ , while the  $(1 - L^4)$ -filter would have a dampening effect for  $\lambda < \pi/12$  and an intensifying effect for  $\pi/12 < \lambda < \pi/4$ .

To assess the possible consequences of using either the  $(1 - L)$ -filter or the  $(1 - L^4)$ -filter in an empirical analysis of financial cycles in growth rates, it is instructive to state that the  $(1 - L^4)$ -filter can be decomposed as follows

$$(1 - L^4) = (1 - L)(1 + L + L^2 + L^3). \quad (18)$$

According to (18), the  $(1 - L^4)$ -filter is equivalent to taking first differences and then forming a moving average of the remainder. Fig. 13 shows the transfer functions of both components of the  $(1 - L^4)$ -filter. The transfer function of the  $(1 + L + L^2 + L^3)$ -filter reflects what is done to the data on top of forming first differences. Note that the transfer function of the  $(1 + L + L^2 + L^3)$ -filter

is quite flat in the gray region which reflects the financial cycle area.

Let us now make the assumption that all (or most) of the true spectral mass of the cyclical component of  $y_t$  is located at longer cycles. If that was actually the case, then, compared to taking first differences, the  $(1 - L^4)$ -filter scales up the true spectrum by a roughly constant number. However, when forming the spectral densities by dividing through the area below the spectrum, the scaling effect becomes irrelevant. Therefore, given all spectral mass is truly located at very low frequencies, the  $(1 - L^4)$ -filter and the  $(1 - L)$ -filter should deliver similar spectral densities. Since we have shown in Section 3.3.5, that for quarterly growth rates of credit and housing the spectral densities are indeed very similar to those obtained from the univariate analysis of yearly growth rates, we conclude that these long cycles are in fact present in the true data generating process.

### B.3. Comparison of indirect and direct spectrum estimation

A direct non-parametric estimation of a spectrum produces generally less precise estimates than the parametric (indirect) approach.<sup>19</sup> In order to get consistent estimates, one has to use a kernel estimator. Thus, the estimates depend on the specific kernel and its bandwidth parameter  $M$ , implying that a large amount of observations is required to have the necessary degrees of freedom. For instance, if one estimates the spectrum from  $T$  observations by

<sup>19</sup> Two interesting applications can be found in Aikman et al. (2015) and Schüller et al. (2015). Aikman et al. (2015) directly estimate the spectral density for UK real loan growth using yearly data from 1880 to 2008 to analyze both the business and the financial cycle, while Schüller et al. (2015) also directly estimate spectral measures to characterize financial cycles across Europe.

transforming the first  $M$  estimated autocovariances, one has  $C \cdot \frac{T}{M}$  degrees of freedom. The constant  $C$  is kernel-specific and usually takes on values of around 3. A small value of  $M$  decreases the variance but increases the bias of the estimator. Having in mind the data in our empirical analysis, let us consider a sample size of  $T = 100$ . In that case, a reasonable choice for  $M$  may be 20, leading to only about 15 degrees of freedom. In contrast, the indirect estimation approach, starting from an ARMA model with e.g. 5 parameters, leaves us with 95 degrees of freedom and hence implies a strong efficiency gain.

An important additional consequence of this reduction in estimation uncertainty is that it facilitates to study possible changes in the spectrum across different time periods. Since the indirect spectrum estimation is based on time domain estimates, it allows the use of conventional break point tests, as e.g. the Chow test, or impulse indicator saturation to endogenously determine breaks.

When estimating spectral densities directly, comparable structural break tests are also available in the frequency domain, see Ahmed et al. (2004) and Pancrazi (2015). Their advantage is that they allow to analyze breaks of (integrated) spectral densities in any frequency range of interest. Generally, however, they are not as precise as tests in the time domain since they are based on a non-parametric approach. During the further procedure, this paper also uses frequency-wise testing to address a number of specific questions, but our testing relies on the indirectly estimated spectra and their bootstrap standard errors.

#### B.4. Bootstrap inference

To statistically assess the characteristics of the financial cycle across variables and different sample periods, the inference in the empirical part is based on bootstrap methods.<sup>20</sup> We apply the following bootstrap procedure, see e.g. Benkwitz et al. (2001).

1. Estimate the parameters of the ARMA model in Eq. (1).
2. Generate bootstrap residuals  $u_1^*, \dots, u_T^*$  by randomly drawing with replacement from the set of estimated residuals.
3. Construct a bootstrap time series recursively using the estimated parameters from step 1 and the bootstrap residuals from step 2.
4. Reestimate the parameters from the data generated in step 3.
5. Repeat step 2 to step 4 5000 times.
6. From the bootstrap distributions of the statistics of interest, e.g. cycle length, amplitude etc., we compute the standard errors and the corresponding 95% confidence intervals.

#### B.5. Estimation example

In order to provide an example of how we use the estimated ARMA models to obtain the spectral densities, consider the following process of US real GDP growth during the pre-1985 period with  $t$ -values in parentheses (cf. C.1) and the notation as in Eq. (1):

$$y_t = 0.002 + 1.144y_{t-1} - 0.224y_{t-3} - 0.985u_{t-4} + u_t.$$

(2.97)      (19.21)      (-4.37)      (-42.04)

These estimates are applied to Eq. (3) to calculate the corresponding spectrum as

$$f_y(\lambda) = \frac{|1 - 0.985 e^{-i4\lambda}|^2}{|1 - 1.144 e^{-i\lambda} + 0.224 e^{-i3\lambda}|^2} \frac{\sigma^2}{2\pi},$$

with  $e^{-i\lambda} = \cos(\lambda) - i \cdot \sin(\lambda)$  by Euler's relation and  $i^2 = -1$ . According to this procedure, we calculated the spectra for all countries and sample periods under consideration. Dividing  $f_y(\lambda)$  by the variance of  $y_t$  yields the spectral densities.

### Appendix C. Detailed time domain estimation results

#### C.1. Estimated ARMA models

(Tables 11–13)

**Table 11**  
ARMA models for the US.

| Parameters  | GDP               |                   |                   | Credit            |                   |                   | Credit to GDP     |                   |                   | Housing           |                   |                   | Equity            |                   |                   |
|-------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|             | Total             | Pre               | Post              | Total             | Pre               | Post              | Total             | Pre               | Post              | Total             | Pre               | Post              | Total             | Pre               | Post              |
| const       | 0.028<br>(18.7)   | 0.031<br>(21.3)   | 0.022<br>(2.95)   | 0.039<br>(24.2)   | 0.044<br>(5.10)   | 0.038<br>(10.5)   | -1.180<br>(-0.50) | 0.239<br>(1.40)   | -1.795<br>(-0.43) | 0.009<br>(4.15)   | 0.001<br>(0.14)   | 0.010<br>(2.48)   | 0.050<br>(2.04)   | -0.049<br>(-5.77) | 0.043<br>(3.25)   |
| AR(1)       | 1.202<br>(29.9)   | 1.144<br>(19.2)   | 1.318<br>(27.1)   | 1.185<br>(79.7)   | 1.237<br>(19.6)   | 1.170<br>(47.3)   | 1.251<br>(28.8)   | 0.873<br>(18.4)   | 1.381<br>(35.8)   | 1.373<br>(18.7)   | 1.130<br>(17.2)   | 1.664<br>(18.5)   | 1.350<br>(20.8)   | 1.302<br>(13.5)   | 1.285<br>(14.3)   |
| AR(2)       |                   |                   |                   |                   |                   |                   |                   |                   |                   | -0.332<br>(-2.58) |                   | -0.827<br>(-4.66) | -0.365<br>(-5.64) | -0.410<br>(-4.33) | -0.333<br>(-3.77) |
| AR(3)       | -0.230<br>(-5.76) | -0.224<br>(-4.37) | -0.328<br>(-6.64) |                   | -0.346<br>(-5.69) |                   | -0.600<br>(-6.49) |                   | -0.402<br>(-10.3) | 0.270<br>(2.10)   |                   | 0.484<br>(2.68)   |                   |                   |                   |
| AR(4)       |                   |                   |                   |                   |                   |                   | 0.939<br>(62.68)  |                   |                   | -0.342<br>(-4.64) | -0.201<br>(-3.28) | -0.345<br>(-3.69) |                   |                   |                   |
| AR(5)       |                   |                   |                   | -0.222<br>(-14.9) |                   | -0.201<br>(-7.92) | -0.611<br>(-10.2) |                   |                   |                   |                   |                   |                   |                   |                   |
| MA(3)       |                   |                   |                   |                   | 0.403<br>(3.68)   |                   | 0.360<br>(3.95)   |                   |                   |                   |                   |                   |                   |                   |                   |
| MA(4)       | -0.978<br>(-92.2) | -0.985<br>(-42.0) | -1.144<br>(-18.1) | -0.904<br>(-29.5) | -0.374<br>(-3.75) | -0.929<br>(-31.6) | -0.429<br>(-5.65) | 0.212<br>(2.76)   |                   | -0.973<br>(-37.2) | -0.983<br>(-29.5) | -0.932<br>(-25.9) | -0.977<br>(-65.2) | -0.943<br>(-47.2) | -0.972<br>(-60.5) |
| MA(5)       |                   |                   |                   |                   | 0.241<br>(2.21)   | 0.487<br>(5.95)   |                   | -0.516<br>(-8.47) |                   |                   |                   |                   |                   |                   |                   |
| MA(6)       |                   |                   |                   |                   | 0.302<br>(2.95)   |                   | -0.185<br>(-2.75) |                   |                   |                   |                   |                   |                   |                   |                   |
| MA(7)       |                   |                   |                   |                   |                   |                   |                   | -0.429<br>(-5.87) |                   |                   |                   |                   |                   |                   |                   |
| MA(12)      |                   |                   | 0.182<br>(3.02)   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |
| diagnostics |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |
| Chow        | 3.37<br>(0.01)    |                   |                   | 1.74<br>(0.14)    |                   |                   | 3.66<br>(0.00)    |                   |                   | 3.97<br>(0.00)    |                   |                   | 3.38<br>(0.01)    |                   |                   |
| LM(4)       | 0.75              | 0.89              | 0.35              | 0.82              | 0.16              | 0.21              | 0.47              | 0.62              | 0.28              | 0.14              | 0.69              | 0.83              | 0.94              | 0.51              | 0.99              |
| LM(8)       | 0.36              | 0.46              | 0.42              | 0.53              | 0.38              | 0.14              | 0.79              | 0.94              | 0.54              | 0.11              | 0.63              | 0.69              | 0.81              | 0.44              | 0.99              |
| LM(12)      | 0.13              | 0.34              | 0.17              | 0.22              | 0.44              | 0.35              | 0.57              | 0.68              | 0.63              | 0.14              | 0.57              | 0.56              | 0.71              | 0.26              | 0.93              |

Notes: The terms *pre* and *post* refer to the sample periods 1960Q1 until 1984Q4 and 1985Q1 until 2013Q4, respectively. *Total* denotes the full sample period. Below the parameter estimates  $t$ -values are given in parentheses. Chow tests with  $p$ -values in parentheses refer to a break in parameters at 1985Q1. For LM( $k$ ) tests of no autocorrelation up to order  $k$  the table shows  $p$ -values.

**Table 12**  
ARMA models for the UK.

| Parameters  | GDP                |                   |                   | Credit            |                   |                   | Credit to GDP     |                   |                   | Housing           |                   |                   | Equity            |                   |                   |
|-------------|--------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|             | Total              | Pre               | Post              | Total             | Pre               | Post              | Total             | Pre               | Post              | Total             | Pre               | Post              | Total             | Pre               | Post              |
| const       | 0.024<br>(23.3)    | 0.021<br>(5.95)   | 0.023<br>(10.8)   | 0.057<br>(12.7)   | 0.043<br>(2.71)   | 0.037<br>(3.71)   | −4.367<br>(−2.87) | −0.094<br>(−0.07) | −1.538<br>(−0.37) | 0.037<br>(7.24)   | 0.001<br>(0.16)   | 0.039<br>(4.64)   | 0.032<br>(3.03)   | −0.024<br>(−0.83) | 0.023<br>(3.05)   |
| AR(1)       | 1.101<br>(32.0)    | 0.941<br>(27.2)   | 1.464<br>(17.2)   | 1.113<br>(36.4)   | 1.174<br>(39.6)   | 1.060<br>(19.6)   | 1.099<br>(37.3)   | 1.046<br>(17.8)   | 1.151<br>(36.0)   | 1.480<br>(36.9)   | 1.321<br>(33.0)   | 1.573<br>(17.1)   | 1.309<br>(18.0)   | 1.372<br>(11.8)   | 1.206<br>(13.3)   |
| AR(2)       |                    |                   | −0.291<br>(−1.98) |                   |                   |                   |                   |                   |                   |                   |                   | −0.291<br>(−1.65) | −0.400<br>(−3.37) | −0.571<br>(−3.05) | −0.289<br>(−3.32) |
| AR(3)       |                    |                   | −0.198<br>(−2.33) |                   |                   |                   |                   |                   |                   | −0.596<br>(−8.03) | −0.446<br>(−12.1) | −0.300<br>(−3.17) | 0.166<br>(1.37)   | 0.446<br>(2.32)   |                   |
| AR(4)       | −0.135<br>(−3.82)  |                   |                   | 0.267<br>(2.28)   |                   |                   | 0.479<br>(4.43)   |                   |                   |                   |                   |                   | −0.235<br>(−1.98) | −0.596<br>(−3.19) |                   |
| AR(5)       |                    |                   |                   | −0.400<br>(−4.00) | −0.192<br>(−6.34) | −0.137<br>(−2.29) | −0.589<br>(−7.04) | −0.120<br>(−2.01) | −0.192<br>(−5.99) | 0.095<br>(2.27)   |                   |                   | 0.123<br>(1.70)   | 0.280<br>(2.40)   |                   |
| MA(4)       | −0.922<br>(−113.6) | −0.946<br>(−34.9) | −0.986<br>(−81.8) | −1.148<br>(−14.9) | −0.954<br>(−39.2) | −0.930<br>(−30.4) | −0.474<br>(−5.33) |                   |                   | −1.120<br>(−12.3) | −0.956<br>(−41.1) | −1.261<br>(−12.6) | −0.979<br>(−74.5) | −0.939<br>(−30.1) | −0.979<br>(−38.6) |
| MA(6)       |                    |                   |                   |                   |                   |                   | −0.229<br>(−3.47) |                   | −0.246<br>(−2.42) |                   |                   |                   |                   |                   |                   |
| MA(8)       |                    |                   |                   |                   |                   |                   | −0.237<br>(−3.40) | 0.150<br>(3.02)   |                   | 0.156<br>(1.75)   |                   | 0.317<br>(3.33)   |                   |                   |                   |
| MA(12)      |                    |                   |                   | 0.174<br>(2.41)   |                   |                   |                   | −0.832<br>(−28.0) |                   |                   |                   |                   |                   |                   |                   |
| diagnostics |                    |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |
| Chow        | 1.86<br>(0.12)     |                   |                   | 5.54<br>(0.00)    |                   |                   | 1.92<br>(0.07)    |                   |                   | 3.24<br>(0.01)    |                   |                   | 1.90<br>(0.07)    |                   |                   |
| LM(4)       | 0.83               | 0.62              | 0.77              | 0.88              | 0.74              | 0.44              | 0.83              | 0.75              | 0.72              | 0.15              | 0.98              | 0.99              | 0.95              | 0.40              | 0.70              |
| LM(8)       | 0.39               | 0.39              | 0.85              | 0.41              | 0.74              | 0.78              | 0.83              | 0.81              | 0.52              | 0.18              | 0.99              | 0.57              | 0.85              | 0.82              | 0.92              |
| LM(12)      | 0.43               | 0.40              | 0.56              | 0.51              | 0.81              | 0.84              | 0.71              | 0.31              | 0.36              | 0.30              | 0.99              | 0.87              | 0.94              | 0.94              | 0.98              |

Notes: The terms *pre* and *post* refer to the sample periods 1960Q1 until 1984Q4 and 1985Q1 until 2013Q4, respectively. *Total* denotes the full sample period. Below the parameter estimates *t*-values are given in parentheses. Chow tests with *p*-values in parentheses refer to a break in parameters at 1985Q1. For LM(*k*) tests of no autocorrelation up to order *k* the table shows *p*-values.

**Table 13**  
ARMA models for Germany.

| Parameters  | GDP               |                   |                   | Credit            |                   |                   | Credit to GDP   |                   |                   | Housing           |                   |                   | Equity            |                   |                   |
|-------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-----------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|             | Total             | Pre               | Post              | Total             | Pre               | Post              | Total           | Pre               | Post              | Total             | Pre               | Post              | Total             | Pre               | Post              |
| const       | 0.017<br>(8.59)   | 0.022<br>(14.1)   | 0.013<br>(14.2)   | −0.159<br>(−0.16) | 0.035<br>(15.0)   | −0.047<br>(−0.83) | –               | −2.168<br>(−0.71) | −0.933<br>(−0.10) | −0.005<br>(−1.93) | −0.001<br>(−0.30) | −0.004<br>(−0.31) | 0.038<br>(5.12)   | 0.065<br>(1.66)   | 0.029<br>(2.06)   |
| AR(1)       | 1.032<br>(24.7)   | 0.820<br>(12.2)   | 1.201<br>(12.4)   | 1.091<br>(28.2)   | 1.159<br>(16.8)   | 1.018<br>(22.3)   | 0.999<br>(55.4) | 0.586<br>(5.11)   | 1.094<br>(19.7)   | 1.545<br>(19.9)   | 1.769<br>(23.0)   | 1.440<br>(14.0)   | 1.233<br>(16.6)   | 1.076<br>(16.9)   | 1.291<br>(13.2)   |
| AR(2)       |                   |                   | −0.363<br>(−3.80) |                   |                   |                   |                 | 0.298<br>(2.53)   |                   | −0.429<br>(−3.06) | −0.811<br>(−10.7) | −0.337<br>(−2.54) | −0.307<br>(−4.17) |                   | −0.386<br>(3.97)  |
| AR(3)       |                   |                   |                   |                   | −0.223<br>(−3.75) |                   |                 |                   |                   | −0.139<br>(−1.77) |                   |                   |                   |                   |                   |
| AR(4)       | −0.072<br>(−1.72) |                   |                   | −0.092<br>(−2.37) |                   | −0.725<br>(−8.61) |                 |                   | −0.518<br>(−3.09) |                   |                   |                   |                   | −0.136<br>(−2.16) |                   |
| AR(5)       |                   |                   |                   |                   |                   | 0.688<br>(9.74)   |                 |                   | 0.396<br>(2.39)   |                   |                   | −0.116<br>(−2.80) |                   |                   |                   |
| MA(4)       | −0.751<br>(−9.90) | −0.615<br>(−5.34) | −0.782<br>(−6.94) | −0.647<br>(−8.86) | −0.912<br>(−28.6) |                   | 0.402<br>(5.74) | 0.520<br>(5.11)   | 0.764<br>(5.93)   | −0.736<br>(−10.9) | −0.751<br>(−7.59) | −0.905<br>(−24.5) | −0.969<br>(−74.5) | −0.923<br>(−21.3) | −0.957<br>(−45.0) |
| MA(8)       | −0.228<br>(−3.03) | −0.302<br>(−2.68) | −0.182<br>(−1.63) | −0.239<br>(−3.18) |                   | −0.907<br>(−38.2) |                 | 0.814<br>(19.3)   |                   |                   |                   |                   |                   |                   |                   |
| MA(12)      |                   |                   |                   |                   |                   |                   |                 | 0.563<br>(6.28)   |                   | −0.198<br>(−3.09) | −0.183<br>(−1.87) |                   |                   |                   |                   |
| diagnostics |                   |                   |                   |                   |                   |                   |                 |                   |                   |                   |                   |                   |                   |                   |                   |
| Chow        | 3.63<br>(0.00)    |                   |                   | 3.68<br>(0.00)    |                   |                   | 1.74<br>(0.18)  |                   |                   | 1.22<br>(0.30)    |                   |                   | 0.30<br>(0.88)    |                   |                   |
| LM(4)       | 0.96              | 0.80              | 0.15              | 0.88              | 0.66              | 0.32              | 0.43            | 0.76              | 0.83              | 0.57              | 0.74              | 0.58              | 0.57              | 0.31              | 0.86              |
| LM(8)       | 0.28              | 0.90              | 0.58              | 0.20              | 0.86              | 0.19              | 0.43            | 0.40              | 0.92              | 0.59              | 0.95              | 0.75              | 0.89              | 0.73              | 0.87              |
| LM(12)      | 0.42              | 0.97              | 0.86              | 0.38              | 0.92              | 0.31              | 0.54            | 0.50              | 0.59              | 0.76              | 0.81              | 0.81              | 0.78              | 0.57              | 0.80              |

Notes: The terms *pre* and *post* refer to the sample periods 1970Q1 until 1990Q1 and 1990Q2 until 2013Q4, respectively. *Total* denotes the full sample period. Below the parameter estimates *t*-values are given in parentheses. Chow tests with *p*-values in parentheses refer to a break in parameters at 1990Q2. For LM(*k*) tests of no autocorrelation up to order *k* the table shows *p*-values.

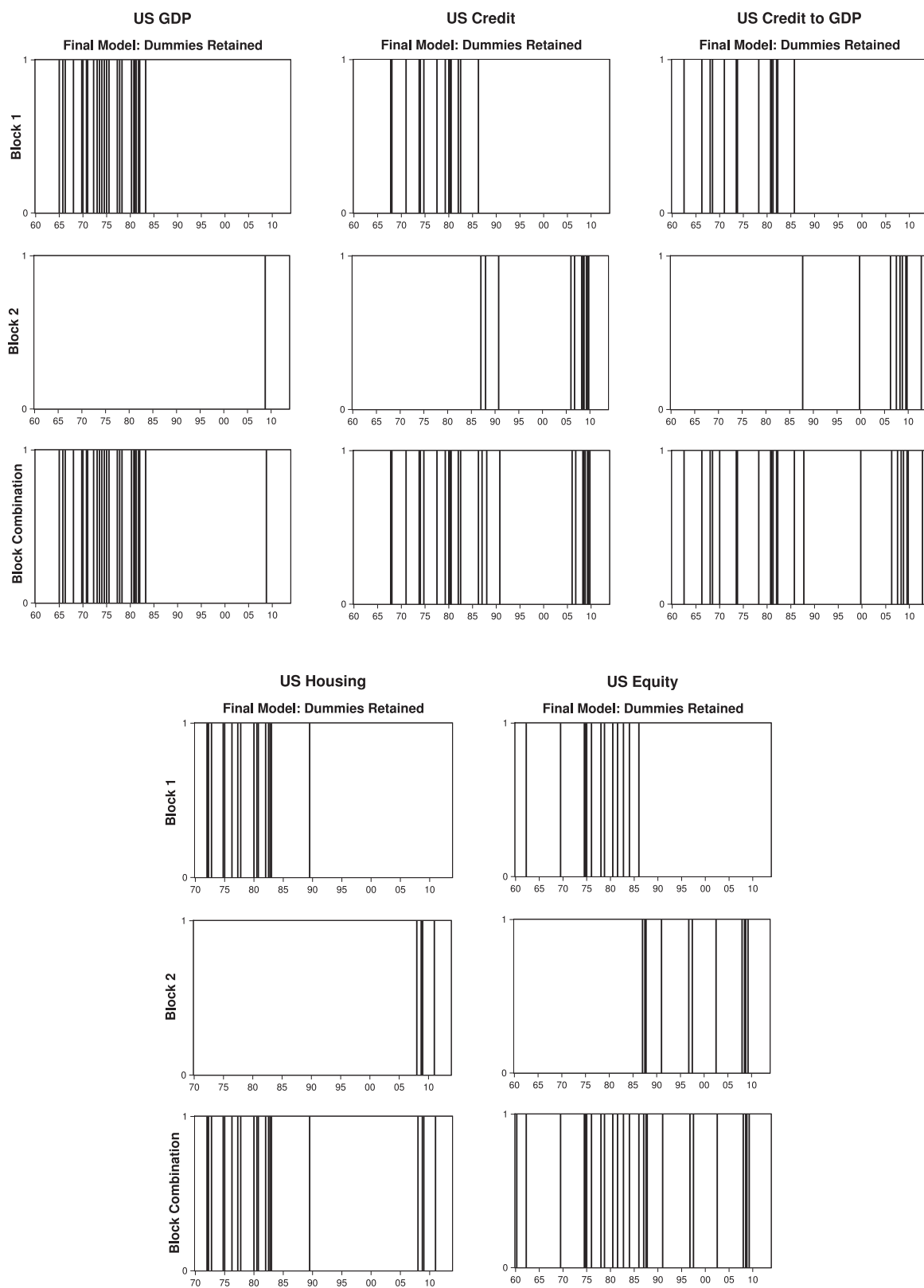
## C.2. Testing for breaks: Impulse indicator saturation

As outlined in Section 3.1 on the data, we want to analyze possible changes in the characteristics of the financial cycle. Thereby, we follow the literature, which commonly specifies the break at 1985Q1, arguing that around this time the process of financial liberalization around the world started to speed up considerably (see e.g. Claessens et al., 2011, 2012 and Drehmann et al., 2012). Chow tests provide some support for the break date 1985Q1; see test results in C.1.

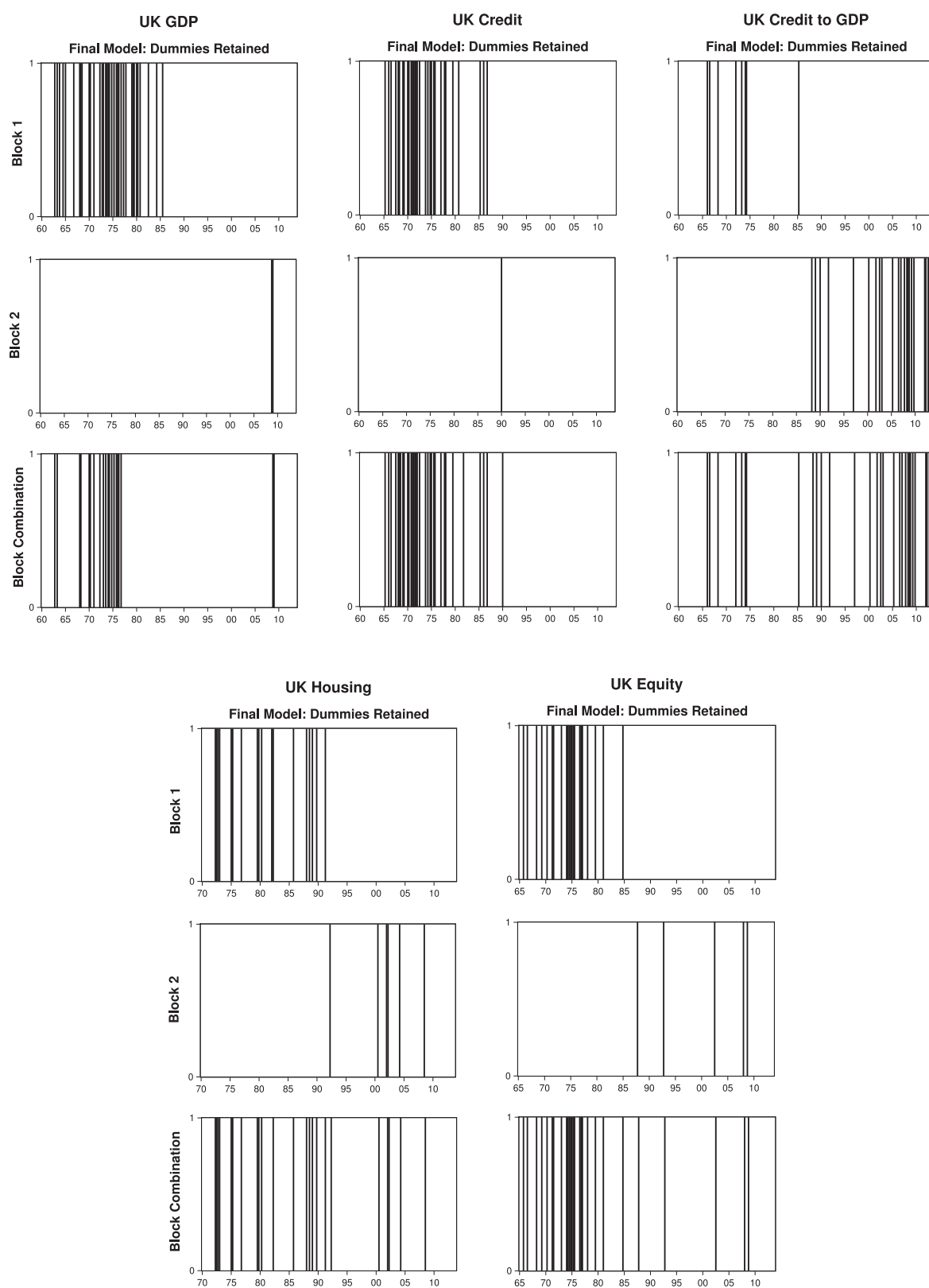
To provide additional statistical support without a priori specifying a break date (as it is necessary for the Chow test), we conducted an impulse indicator saturation analysis. Following Hendry (2011) and Ericsson (2013) (see also the references therein), we use the split-half approach. That is, we include impulse dummies for all data points of the first half of the sam-

ple and estimate the model over the full sample. From the estimation results, we identify the significant dummies (see Block 1 in Figs. 14–16). Similarly, we then repeat the estimation including dummies only for the second half of the sample (see Block 2). Thereafter, we estimate the model including all dummies which were significant in first two steps of the test procedure. Finally, we retain all dummies being significant at the 10%-level (see Block Combination).

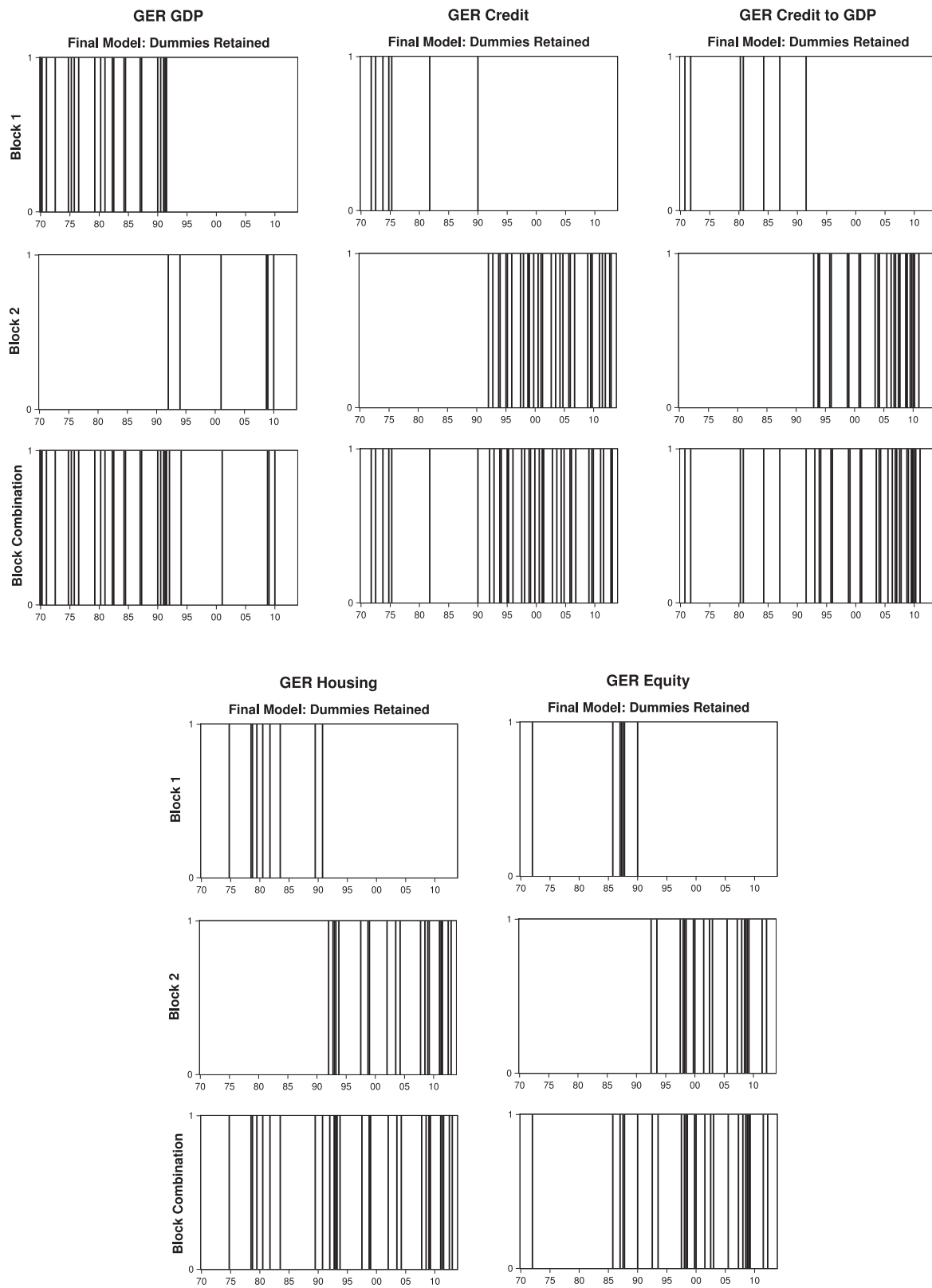
From the test results in Figs. 14–16, it can be seen that for most variables a considerable amount of dummies remains significant in one half of the sample while in the other half only a few dummies survive. Often the few surviving dummies are found in the second sample half and only pick up outliers during the financial crisis. While the test procedure does not provide specific break dates, it gives in most cases fairly strong indication that the model parameters are subject to structural change. In particular, the results in-



**Fig. 14.** Endogenous Structural Break Point Search for the US. Note: The figure shows the results from the Impulse Indicator Saturation procedure using the split-half approach.



**Fig. 15.** Endogenous Structural Break Point Search for the UK. *Note:* The figure shows the results from the Impulse Indicator Saturation procedure using the split-half approach.



**Fig. 16.** Endogenous Structural Break Point Search for the GER. *Note:* The figure shows the results from the Impulse Indicator Saturation procedure using the split-half approach.

dicate a break around mid to late 1980s for US and UK and around 1990 for Germany.

Note that the IIS procedure is quite general. It is possible to interpret many existing tests as special cases of IIS, as e.g. recursive estimation, rolling regression, the [Chow \(1960\)](#) test, the [Andrews \(1993\)](#) endogenous break point test, the [Bai and Perron \(1998\)](#) multiple breakpoint test and the tests of extended constancy in [Ericsson et al. \(1998\)](#).

## References

- A'Hearn, B., Woitek, U., 2001. More international evidence on the historical properties of business cycles. *J. Monet. Econ.* 47 (2), 321–346. doi:10.1016/S0304-3932(01)00045-9.
- Ahmed, S., Levin, A., Wilson, B.A., 2004. Recent U.S. macroeconomic stability: good policies, good practices, or good luck? *Rev. Econ. Stat.* 86 (3), 824–832.
- Aikman, D., Haldane, A.G., Nelson, B.D., 2015. Curbing the credit cycle. *Econ. J.* 125 (585), 1072–1109. doi:10.1111/econj.12113.
- Andrews, D.W., 1993. Tests for parameter instability and structural change with unknown change point. *Econometrica* 61 (4), 821–856.
- Ardila, D., Sornette, D., 2016. Dating the financial cycle with uncertainty estimates: a wavelet proposition. *Finance Res. Lett.* 19, 298–304.
- Bai, J., Perron, P., 1998. Estimating and testing linear models with multiple structural changes. *Econometrica* 66 (1), 47–78.
- Barbone Gonzalez, R., Lima, J., Marinho, L., 2015. Business and Financial Cycles: an estimation of cycles' length focusing on Macropudential Policy. Working Paper 385. Banco Central do Brasil.
- Baxter, M., King, R.G., 1999. Measuring business cycles: approximate band-pass filters for economic time series. *Rev. Econ. Stat.* 81, 575–593.
- Beaudry, P., Galizia, D., Portier, F., 2016. Putting the Cycle Back into Business Cycle Analysis. Working Paper 22825. National Bureau of Economic Research doi:10.3386/w22825.
- Bekaert, G., Hoerova, M., Lo Duca, M., 2013. Risk, uncertainty, and monetary policy. *J. Monet. Econ.* 60 (7), 771–788.
- Benkwitz, A., Lütkepohl, H., Wolters, J., 2001. Comparison of bootstrap confidence intervals for impulse responses of german monetary systems. *Macroecon. Dyn.* 5 (01), 81–100.
- Borio, C., 2014. The financial cycle and macroeconomics: what have we learnt? *J. Bank. Finance* 45 (395), 182–98.
- Borio, C., Zhu, H., 2012. Capital regulation, risk-taking and monetary policy: a missing link in the transmission mechanism? *J. Financ. Stud.* 8 (4), 236–251.
- Breitung, J., Candelon, B., 2006. Testing for short- and long-run causality: a frequency-domain approach. *J. Econom.* 132 (2), 363–378. doi:10.1016/j.jeconom.2005.02.004.
- Breitung, J., Schreiber, S., 2017. Assessing causality and delay within a frequency band. *Econom. Stat.* doi:10.1016/j.ecosta.2017.04.005. In Press.
- Bruno, V., Shin, H.S., 2015. Capital flows and the risk-taking channel of monetary policy. *J. Monet. Econ.* 71, 119–132.
- Bry, G., Boschan, C., 1971. Cyclical Analysis of Time Series: Selected Procedures and Computer Programs. NBER.
- Chow, G.C., 1960. Tests of equality between sets of coefficients in two linear regressions. *Econometrica* 28 (3), 591–605.
- Christiano, L.J., Fitzgerald, T.J., 2003. The band pass filter. *Int. Econ. Rev.* 44 (2), 435–465.
- Claessens, S., Kose, M.A., Terrones, M.E., 2011. Financial Cycles: What? How? When? In: Clarida, R., Giavazzi, F. (Eds.), NBER International Seminar on Macroeconomics, 7. University of Chicago Press, pp. 303–344.
- Claessens, S., Kose, M.A., Terrones, M.E., 2012. How do business and financial cycles interact? *J. Int. Econ.* 97, 178–190.
- Domanski, D., Ng, T., 2011. Getting effective macroprudential policy on the road: eight propositions. BIS Paper 60. Bank for International Settlements.
- Drehmann, M., Borio, C., Tsatsaronis, K., 2012. Characterizing the final cycle: Don't lose sight of the medium term!. BIS Working Paper 380. Bank for International Settlements.
- ECB, 2014. Financial Stability Report. European Central Bank, November 2014.
- English, W., Tsatsaronis, K., Zoli, E., 2005. Assessing the predictive power of measures of financial conditions for macroeconomic variables. BIS Papers 22, 228–252.
- Ericsson, N.R., 2013. How biased are US government forecasts of the federal debt? Draft, Board of Governors of the Federal Reserve System, Washington, DC.
- Ericsson, N.R., Hendry, D.F., Prestwich, K.M., 1998. The demand for broad money in the United Kingdom, 1878–1993. *Scand. J. Econ.* 100 (1), 289–324.
- ESRB, 2014. Recommendation of the European Systemic Risk Board on guidance for setting countercyclical buffer rates. Off. J. Eur. Union, C 293/01.
- Fisher, I., 1933. The debt-deflation theory of great depressions. *Econometrica* 1, 337–357.
- Galati, G., Hindrayanto, I., Koopman, S.J., Vlekke, M., 2016. Measuring financial cycles with a model-based filter: Empirical evidence for the United States and the euro area. Working Paper 495. De Nederlandsche Bank, Amsterdam.
- Geanakoplos, J., 2010. The leverage cycle. NBER Macroecon. Annu. 24 (1), 1–66.
- Geanakoplos, J., Axtell, R., Farmer, D.J., Howitt, P., Conlee, B., Goldstein, J., Hendrey, M., Palmer, N.M., Yang, C.-Y., 2012. Getting at systemic risk via an agent-based model of the housing market. *Am. Econ. Rev.* 102 (3), 53–58.
- Geweke, J.F., 1982. Measurement of linear dependence and feedback between multiple time series. *J. Am. Stat. Assoc.* 77 (378), 304–313. doi:10.1080/01621459.1982.10477803.
- Geweke, J.F., 1984. Measures of conditional linear dependence and feedback between time series. *J. Am. Stat. Assoc.* 79 (388), 907–915. doi:10.1080/01621459.1984.10477110.
- Hamilton, J.D., 1994. Time Series Analysis. Princeton University Press, Princeton.
- Harding, D., Pagan, A., 2002. Dissecting the cycle: a methodological investigation. *J. Monet. Econ.* 49, 365–381.
- Harvey, A.C., Trimbur, T.M., 2003. General model-based filters for extracting cycles and trends in economic time series. *Rev. Econ. Stat.* 85 (2), 244–255.
- Hatzius, J., Hooper, P., Mishkin, F., Schoenholtz, K., Watson, M., 2010. Financial Conditions Indexes: A Fresh Look After the Financial Crisis. Working Paper 16150. National Bureau of Economic Research, Cambridge, MA.
- Hayek, F.A., 1933. Monetary Theory and the Trade Cycle. Clifton, New Jersey: Augustus M. Kelly reprint 1966..
- Hendry, D., 2011. Justifying empirical macro-econometric evidence. In: Journal of Economic Surveys, Online 25th Anniversary Conference, November.
- Hodrick, R.J., Prescott, E.C., 1997. Postwar u.s. business cycles: an empirical investigation. *J. Money Credit Bank.* 29, 1–16.
- Johansen, S., 1995. Likelihood-based Inference in Cointegrated Vector Autoregressive Models. Oxford University Press.
- Jordà, Ò., Schularick, M., Taylor, A.M., 2017. Macrofinancial history and the new business cycle facts. NBER Macroecon. Annu. 31 (1), 213–263.
- Juselius, M., Borio, C., Disyatat, P., Drehmann, M., 2016. Monetary policy, the financial cycle and ultralow interest rates. BIS working paper 569. Bank for International Settlements.
- Juselius, M., Drehmann, M., 2016. Leverage dynamics and the burden of debt. Bank of Finland Research Discussion Paper No. 3.
- Keynes, J.M., 1936. The General Theory of Employment, Interest and Money. MacMillan, New York.
- Khinchine, A., 1934. Korrelationstheorie der stationären stochastischen Prozesse. *Mathematische Annalen* 109 (1), 604–615.
- King, R.G., Rebelo, S.T., 1993. Low frequency filtering and real business cycles. *J. Econ. Dyn. Control* 17 (1), 207–231.
- König, H., Wolters, J., 1972. Zum Problem langfristiger Wachstumszyklen: Eine Spektralanalyse der englischen Entwicklung von 1700–1913. *Zeitschrift für die gesamte Staatswissenschaft/J. Inst. Theoret. Econ.* 72–96.
- Koopman, S.J., Lucas, A., 2005. Business and default cycles for credit risk. *J. Appl. Econom.* 20 (2), 311–323.
- Maravall, A., Mathis, A., 1994. Encompassing univariate models in multivariate time series: a case study. *J. Econom.* 61 (2), 197–233.
- Menden, C., Proaño, C.R., 2017. Dissecting the financial cycle with dynamic factor models. *Quant. Finance*. Forthcoming. doi: 10.1080/14697688.2017.1357971.
- Minsky, H.P., 1982. Can 'It' Happen Again? Essays on Instability and Finance. M. E. Sharpe, Armonk.
- Miranda-Agrippino, S., Rey, H., 2015. World Asset Markets and the Global Financial Cycle. Working Paper 21722. National Bureau of Economic Research, Cambridge, MA.
- von Mises, L., 1952. The Theory of Money and Credit. Yale University Press.
- Murray, C.J., 2003. Cyclical properties of Baxter-King filtered time series. *Rev. Econ. Stat.* 85 (2), 472–476.
- Ng, T., 2011. The predictive content of financial cycle measures for output fluctuations. *BIS Q. Rev.* June, 53–65.
- Pancrazi, R., 2015. The heterogeneous great moderation. *Eur. Econ. Rev.* 74 (0), 207–228. doi:10.1016/j.euroecorev.2014.12.005.
- Passari, E., Rey, H., 2015. Financial flows and the international monetary system. *Econ. J.* 125.584 (21172), 675–698.
- Rey, H., 2013. Dilemma not trilemma: The global financial cycle and monetary policy independence. In: Economic Policy Symposium. Federal Reserve Bank of Kansas City, Jackson Hole, pp. 285–333.
- Rey, H., 2016. International channels of transmission of monetary policy and the Mundellian trilemma. *IMF Econ. Rev.* 64 (1), 6–35.
- Rünstler, G., Vlekke, M., 2018. Business, housing, and credit cycles. *J. Appl. Econom.* 33 (2), 212–226. doi:10.1002/jae.2604.
- Schularick, M., Taylor, A.M., 2012. Credit booms gone bust: monetary policy, leverage cycles, and financial crises, 1870–2008. *Am. Econ. Rev.* 102 (2), 1029–61.
- Schüler, Y.S., Hiebert, P., Peltonen, T.A., 2015. Characterising the financial cycle: a multivariate and time-varying approach. Working Paper 1846. European Central Bank, Frankfurt am Main.
- Toda, H.Y., Yamamoto, T., 1995. Statistical inferences in vector autoregressions with possibly integrated processes. *J. Econom.* 66, 225–250.
- Verona, F., 2016. Time-frequency characterization of the US financial cycle. *Econ. Lett.* 144, 75–79.
- Wiener, N., 1930. Generalized harmonic analysis. *Acta Mathematica* 55 (1), 117–258.
- de Winter, J., Koopman, S.J., Hindrayanto, I., Chouhan, A., 2017. Modeling the business and financial cycle in a multivariate structural time series model. De Nederlandsche Bank Working Paper No. 573.
- Wolters, J., 1980a. Business cycle stabilization policies in a small econometric model of the FRG. *Eur. Econ. Rev.* 14 (1), 9–43.
- Wolters, J., 1980b. Stochastic Dynamic Properties of Linear Econometric Models. In: Beckmann, M., Künzi, H. (Eds.), Lecture Notes in Economics and Mathematical Systems, 182. Springer Berlin Heidelberg, pp. 1–154.